



User-Oriented Infrastructure Reports: From Data to Story

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ABSTRACT

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Infrastructure condition reports often have lots of technical details but are not always clear or useful for the stakeholders who rely on them. This thesis examined how public sector infrastructure reports can be made more user-oriented, accessible, and relevant through a case study with Vöylävirasto (Finnish Transport Infrastructure Agency, FTIA). A practical reporting template and accompanying guidelines were developed to transform technical road and bridge condition data into concise narratives that answer stakeholder questions and support decision-making.

The study was conducted as a design-based research case, combining iterative prototyping, stakeholder engagement, and international benchmarking. A stakeholder-oriented report template was co-designed with Vöylävirasto staff through cycles of feedback and refinement, structured conceptually by Hevner's Three-Cycle Model. Qualitative interviews, internal document analysis, and past presentation materials were used to identify user needs and communication barriers among internal experts, policymakers, journalists, and citizens. International practices, particularly the UK's ALARM report and Ireland's asset management reporting, were reviewed to adapt effective visual and narrative techniques to the Finnish context. The emerging template and guidelines were aligned with key communication and accessibility requirements drawn from ISO 55001 and the Web Content Accessibility Guidelines to support clarity, consistency, and inclusive use.

The main outcome is a modular, question-based reporting template and a set of practical guidelines that can be adapted for future use. The template organizes information around frequently asked stakeholder questions and employs color-coded visuals, plain-language summaries, glossaries, and annotated figures to reduce jargon and highlight key messages. Internal feedback indicated that these features make complex infrastructure topics easier to communicate both within and outside the organization and help standardize reporting practices. Although the template has not yet been tested with external users, the findings indicate that user-oriented data storytelling can make infrastructure data more meaningful, support better decisions, and strengthen trust in public sector communication. Future work should include broader piloting with external audiences and exploration of digital tools, such as AI-assisted reporting and chatbots, to further enhance accessibility and engagement.

Keywords: User-oriented reporting, infrastructure data, data storytelling, stakeholder needs, accessibility, public sector communication, design-based research, benchmarking, template development, visual communication.

Use of AI in Thesis

I have used AI tools in my thesis:

☐ No

☒ Yes

Names of the tools and versions:

Microsoft 365 Copilot - GPT-5 Chat-based model

Grammarly for Windows: Version 1.2.228.1814

OpenAI ChatGPT using the GPT-4o model

Purpose of the use:

During my thesis process, I used AI to help me improve the clarity and quality of the text I had written myself. I wanted to make my own writing easier to read and more consistent. I used AI like checking grammar, making my explanations and sentence structure better. I would prompt AI to make my drafted text clearer, especially when something felt awkward or repetitive. I used AI to help me compare different ways of wording the same idea so that I could choose the option that fit the academic style and flow of the thesis. I asked AI for suggest headings and the order of topics. However, all the arguments, analysis, and conclusions in the thesis were created independently by me.

ChatGPT was also used to help generate a few illustrative visuals (e.g., example slide/figure mockups) to demonstrate design principles; these figures are marked as AI-generated in their captions, and the content was reviewed and edited by the researcher.

Parts in which AI was used:

- Title and Abstract: I used AI to suggest titles and headings. I used AI to help check the grammar and wording and help me shorten sentences.
- Introduction: AI helped with improving how smoothly the text was read and re-organizing some sentences.
- Literature Review: I used AI to make my long academic paragraphs more readable. I only improved the clarity of paragraphs I had written myself.

- Methodology: I used AI for grammar, sentence structure, and translating a few phrases from Finnish to English.
- Discussion and Conclusion: I used AI for grammar, sentence structure, and translating a few phrases from Finnish to English. In these sections, AI helped me organize long paragraphs and improve readability. I sometimes asked it to suggest alternatives for sentences that were awkward or too repetitive. The actual conclusions are completely my own.

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ABBREVIATIONS

Abbreviation	Full Term / Meaning
AI	Artificial Intelligence
ALARM	Annual Local Authority Road Maintenance (UK report)
DBR	Design-Based Research
FAQ	Frequently Asked Questions
FTIA	Finnish Transport Infrastructure Agency (Väylävirasto)
GDPR	General Data Protection Regulation
IRI	International Roughness Index
ISO	International Organization for Standardization
MB	Maintenance Backlog
NLP	Natural Language Processing
OGD	Open Government Data
PIARC	World Road Association
PPIAF	Public-Private Infrastructure Advisory Facility (World Bank)
ROMA	Road Markings Assessment (NordFoU project)
SAMP	Strategic Asset Management Plan (Ireland)
SFS	Suomen Standardit ry (Finnish Standards Association)
WCAG	Web Content Accessibility Guidelines
WCEAM	World Congress on Engineering Asset Management

1 INTRODUCTION

1.1 Background

Väylävirasto (Finnish Transport Infrastructure Agency, FTIA) is responsible for planning, developing, and maintaining Finland's state-owned transport infrastructure, including roads, railways, and waterways. Väylävirasto works under the Ministry of Transport and Communications. Its job is to make sure that the transport network functions well, supporting people's mobility, safety, and economic competitiveness. Väylävirasto is responsible for Finland's state highway network, which is about 78 000 kilometers in length (Väylävirasto, 2026). This includes more than 13 000 kilometers of main roads, approximately 900 kilometers of motorways, and roughly 50 000 kilometers of paved roads (Väylävirasto, 2026). Finland's entire road network, including municipal and private roads, covers approximately 454 000 kilometers (Väylävirasto, 2026). For waterways, Väylävirasto maintains about 8 300 kilometers of coastal fairways and about 8 000 kilometers of inland fairways, making a total of approximately 16 300 kilometers under its responsibility (Väylävirasto, 2022).

Väylävirasto manages significant public funding and employs staff across Finland to coordinate infrastructure work. Included in Väylävirasto's broad scope is the assessment and maintenance of road and bridge conditions. For example, through organized, structured road and bridge maintenance programs. They also track the condition of these structures and share that information publicly. This data helps them make informed decisions about managing their assets, ensuring that they focus on maintaining and investing in the most critical parts of their infrastructure. Väylävirasto communicates with the public through several channels, such as traffic updates, project communications, and online feedback tools, to keep stakeholders informed about infrastructure and planning.

Väylävirasto's infrastructure reports often include large amounts of technical data, but that does not always mean the report is easy to understand and beneficial to the intended audience. The main reason behind this thesis is a simple but

persistent issue: even when there is plenty of information available about the condition of roads and bridges, it is not always presented in a way that makes sense to different audiences. Whether it is policymakers, journalists, or everyday citizens, each group has its own questions and priorities. This can make funding, maintenance, and safety decisions more complicated, and weaken public trust. This thesis is carried out as a development project, as its purpose is to design, test, and refine a practical reporting solution in close collaboration with Väylävirasto.

1.2 Objective

By structuring the report template around these questions and using data storytelling techniques such as clear visuals, narrative headlines, and plain-language summaries, the thesis demonstrates how technical information can be transformed into meaningful stories. This approach ensures that each section of the report template not only answers a stakeholder's question, but also provides context, highlights trends, and connects data to everyday experiences, making the information more relevant and actionable for all audiences.

The approach is practical: it looks at real-world challenges faced by Väylävirasto, as it tries to communicate complex road and bridge data to a wide range of users. The project was carried out as a case study in close collaboration with Väylävirasto, using an iterative, design-based research process. This meant working directly with staff to co-design a new reporting template. Feedback was gathered at each stage, and improvements were made based on what worked and what did not.

To make sure the solutions were not just theoretical, the thesis also benchmarked international best practices. Reports from the UK and Ireland, for example, showing how clear visuals, modular layouts, and plain language can make technical information accessible to non-experts. These examples provided inspiration and a standard to aim for, helping to shape the design of the new report template for Väylävirasto.

The main objective of this thesis is to develop a user-oriented reporting template that makes infrastructure data more understandable and relevant for different stakeholder groups. The goal of this developmental project is to offer a practical model and set of guidelines that can be reused and adapted in the future. By focusing on stakeholder needs, data storytelling, and international benchmarks, the thesis aims to show how public infrastructure reporting can be clearer and more consistent. In the end, the hope is that better communication will support better decisions and make the context of the data more comprehensible to a wider audience.

The thesis aims to demonstrate how user-oriented design, data storytelling, and international benchmarks can transform complex technical information into clear, meaningful insights that support decision-making. Ultimately, the intent is to improve communication quality, increase transparency, and ensure that infrastructure data is comprehensible to a wide audience.

1.3 Thesis Questions

The main concern of this thesis work is guided by the core research question:

How can public sector infrastructure reports be transformed to maximize clarity, accessibility, and relevance for diverse stakeholder groups through user-oriented design, data storytelling, and the adaptation of international best practices?

To address this broader question, the thesis investigates the following sub-questions:

- Question 1: To what extent does a user-oriented reporting template improve the accessibility and usefulness of infrastructure data for different stakeholder groups (e.g., policymakers, journalists, citizens, and internal staff)?
- Question 2: To what extent does structuring infrastructure reports around frequently asked stakeholder questions enhance the clarity and relevance of the information presented?

- Question 3: To what extent do data storytelling techniques such as clear visuals, narrative headlines, and plain-language summaries facilitate the transformation of technical infrastructure data into meaningful and actionable stories?
- Question 4: To what extent can international effective practices and standards in infrastructure reporting be adapted to improve the clarity, consistency, and accessibility of Finnish public sector reports?

1.4 Limitations

This thesis has several limitations that influenced the scope, findings, and the kinds of conclusions that can be drawn. First, the study focuses on a single case organization, Väylävirasto, which means that the results may not directly apply to other agencies or public institutions. The design process and findings are shaped by Väylävirasto's internal practices, priorities, and available data, so generalizing the results should be done cautiously. Second, the data collection relied only on internal stakeholders, mainly staff members involved in reporting, asset management, communications, and customer service. External users such as citizens, journalists, or policy advisors did not participate directly. This may have narrowed the range of perspectives included in the research and introduced a risk of internal bias. Third, the main output, a reporting template built with mock data has not yet been tested in reporting cycles, so its effectiveness is based on stakeholder expectations rather than measured outcomes. To be transparent, the following section summarizes the key limitations chapter by chapter, explaining how each stage of the research was shaped by practical constraints, methodological choices, and the researcher's capabilities as a student researcher.

1.4.1 Chapter 1: Introduction

The limitation here mainly relates to scoping. Because Vöylävirasto requested a focused development project, the thesis centers on improving one specific communication challenge road and bridge condition reporting. This narrowed the problem definition early on and may have excluded other valuable but less visible needs.

1.4.2 Chapter 2: Literature Review

The literature review combines areas such as data storytelling, accessibility, asset management standards, and international benchmarks. A limitation is that the review depends on published work and publicly available reports, which do not always include detailed evaluations of user comprehension or the internal decision-making processes behind internationally successful reports. Because some countries (e.g., UK, Ireland) publish strong visuals but limited methodology, the review may reflect best practices in form, but not always in process.

1.4.3 Chapter 3: Methodology

The main limitation here is the scope of participants. All interviews were conducted with internal Vöylävirasto staff. This provides depth, but it also risks internal confirmation bias, because staff members share similar working cultures, tools, and assumptions. I did not interview policymakers, journalists, or citizens directly, even though they are important user groups. The DBR approach also focuses on iterative improvement rather than controlled comparison, so the template was shaped by evolving feedback rather than formal testing.

1.4.4 Chapter 4: Findings

The findings reflect what internal staff considered barriers and needs. This means the results may be over-representative of internal frustrations (e.g., inconsistent

templates, reliance on individual expertise) and may under-represent external frustrations (e.g., journalists' deadlines, citizens' difficulty using Suomen Väylät). The visual examples, maps, and charts in the findings were mostly based on mock data and sample visuals, so some insights may differ when real operational data is integrated. In addition, findings about usability or clarity are based on internal feedback only, so the conclusions may be optimistic compared to the experience of external audiences.

1.4.5 Chapter 5: Suggestions

The suggestions chapter includes practical recommendations, but these suggestions have not been piloted, nor validated through official organizational processes. Some proposals (e.g., a Reporting Design Lead, story briefs, data story packs) depend on organizational capacity, staffing, and long-term commitment, factors the research could not test as a student researcher. Recommendations involving AI (e.g., using Copilot for visual generation) come from theoretical literature rather than specific testing inside Väylävirasto because such pilots were outside the thesis scope.

1.4.6 Chapter 6: Conclusion

The broader conclusions reflect internal insights and literature alignment, but because there was no external user testing, the final claims about clarity, usefulness, or accessibility remain theoretical. The study does not measure comprehension, retention, or user satisfaction in real reporting situations. Additionally, the template has not yet gone through accessibility audits or live deployment, so conclusions about accessibility improvements are anticipated rather than verified.

2 REVIEW OF LITERATURE

Before examining themes of data storytelling, visualization, asset management, accessibility, and public-sector communication, it is important to outline why these theoretical areas were chosen as the foundation for this thesis. Infrastructure reporting operates at the intersection of technical data, public communication, and stakeholder engagement, meaning that no single theory is sufficient to address the practical challenges identified in Vöylävirasto's current reporting practices (Schröder et al., 2023; Kosara & Mackinlay, 2013; Segel & Heer, 2010). Therefore, the literature review brings together theories from four complementary domains: (1) data storytelling and narrative visualization, which explain how complex data can be transformed into meaningful, audience-relevant insights (Schröder et al., 2023; Kosara & Mackinlay, 2013; Shao et al., 2024; Oberascher et al., 2023); (2) design-based research and design science, which provide methodological grounding for iterative, user-oriented template development (Barab & Squire, 2004; Hevner, 2007; Kempeneer et al., 2025); (3) public-sector communication, accessibility frameworks, and legal standards, which ensure that reporting practices are inclusive, transparent, and aligned with regulatory requirements (W3C, 2023; European Union, 2019; Laamanen et al., 2022; Kempeneer et al., 2022; Oliveira et al., 2023); and (4) asset management and international infrastructure reporting standards, which anchor the work in the technical and organizational context of transport infrastructure (SFS Finnish Standards, 2025; Asphalt Industry Alliance, 2025; Transport Infrastructure Ireland, 2023; Konstantakos et al., 2019; Hämäläinen, Vaismaa, & Kolisoja, 2025). These theoretical pillars were selected because together they explain what effective communication requires, how stakeholder-oriented solutions can be developed, and why standardized, accessible, and narrative-driven reporting is essential for public agencies responsible for complex technical data (Hevner, 2007; ISO 24495-1:2023; W3C, 2023).

DBR/DSR keeps the process rigorous and iterative (plan → prototype → test → refine), while plain-language and accessibility guidelines make sure the final outputs are clear and usable for all stakeholders (Hevner, 2007; ISO, 2023; W3C, 2023). In practice, the report template is theoretically well designed because it includes concrete choices like descriptive slide titles, short sentences, readable

fonts, and high-contrast color palettes that meet WCAG thresholds, plus alt text for graphics so content works with assistive technologies (ISO, 2023; W3C, 2023). These standards help reduce common barriers like jargon and inconsistent terminology by pushing the presenter to define key terms in plain language and keep layouts predictable across sections (ISO, 2023). Altogether, the methods and standards make it easier to produce reports that can be scaled, reused, and understood not only by technical specialists, but also by policymakers, citizens, journalists, and internal staff, (Hevner, 2007; ISO, 2023; W3C, 2023).

2.1 Data Storytelling

Storytelling is a fundamental human practice used for making sense of the world and sharing knowledge using narrative, character, and emotion (Schröder et al., 2023). With data, these principles convert numbers into compelling, accessible insights, a process called data storytelling (Schröder et al., 2023). Data storytelling is “the art of using language and other communicative abilities to enhance traditional data visualizations,” by providing “context and complements the data visualization” (Kempeneer et al., 2025). A data-driven story combines words, images, and visualizations to transform raw data into meaningful information (Kosara & Mackinlay, 2013). Data storytelling plays a key role in making complex data more accessible and trustworthy through human-centered framing. Data storytelling explains the “why” behind the “what” in visualization; early examples were meant to show and explain rather than just analyze (Oberascher et al., 2023).

Data storytelling combines art and science, as it analyzes data and needs to communicate the message clearly (Oberascher et al., 2023). The researcher would suggest that soft skills such as empathy, creativity, curiosity, and critical thinking are important for effective data storytelling. In addition, understanding stakeholder needs and data literacy levels draws on ideas from various fields, including psychology, economics, and journalism (Oberascher et al., 2023).

2.2 Data Storytelling in the Public Sector

Data storytelling is increasingly used in civic and public sectors, including government reporting, public health, and education (Schröder et al., 2023). Using data storytelling in these contexts can help make complex data accessible to a broader audience (Oberascher et al., 2023). For example, the UK's 2025 Annual Local Authority Road Maintenance report (ALARM) uses color-coded scorecards, regional maps, and annotated charts to communicate funding needs and performance trends to non-technical audiences. As shown in FIGURE 1, the report presents the overall structural condition of local roads in England and Wales using a clear, color-coded visual. This illustration categorizes roads as good, adequate, or poor, helping readers quickly see the proportions. Visualizations like this summarize survey data and emphasize key issues, such as the high percentage needing urgent maintenance (Asphalt Industry Alliance, 2025).

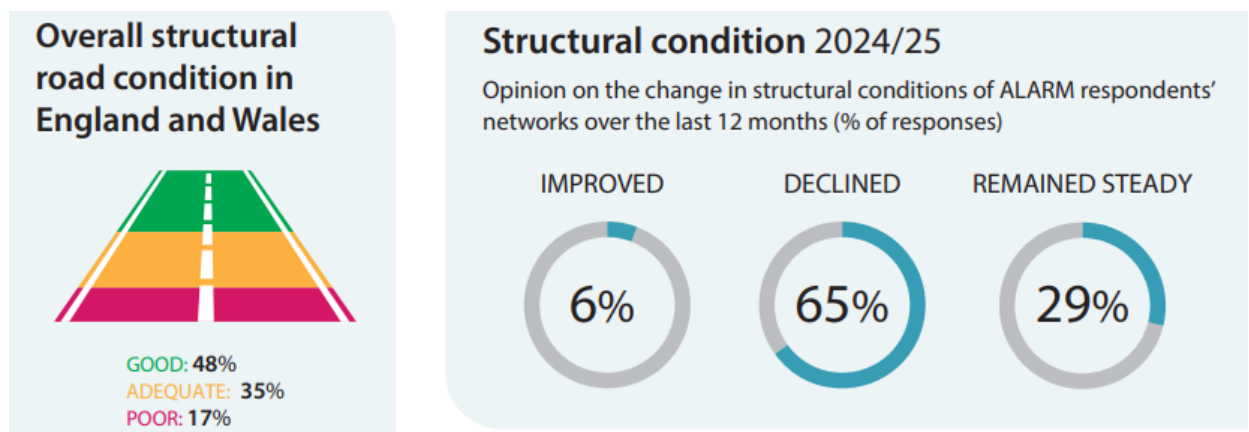


FIGURE 1. Structural condition of local roads in England and Wales, 2024/25. Adapted from ALARM report 2025 (Asphalt Industry Alliance, 2025).

Visual elements such as typography and color clear layouts help guide the audience's attention to the main messages (Schröder et al., 2023). Using color, especially contrast or bold lines like a red line, makes it easy to spot important data (Shao et al., 2024). Additionally, annotations and subheadings help improve clarity in reporting by highlighting key points, providing context, and making information more accessible for non-experts (Kempeneer et al., 2025). Furthermore, subheadings organize information, aid navigation, and boost readability. While color and annotations help readers quickly spot important information, plain text remains essential for deeper understanding. In contrast, too much text or poor design may impede comprehension (Shao et al., 2024).

2.3 Narrative Formats and Cognitive Processing

The structure of a report plays a key role in how the public understands the data. Its clarity and impact are shaped by narrative formats, cognitive processing, and design elements (Schröder et al., 2023). There are many narrative formats including linear, exploratory, and hybrid structures, such as the martini glass model, which begins broadly, narrows to a central argument or focus, and concludes with an open-ended exploration (Kosara & Mackinlay, 2013). For example, starting with a general overview of Finland's road conditions, then focus on a few busy roads that need urgent repairs. Then, it could let stakeholders see local road conditions or upcoming projects in their area. This would help guide the audience with general context, then focus on insights, then allow them to explore further on their own.

The linear narrative format presents data in a controlled, author-guided manner showing change over time and establishing clear cause-and-effect relationships (Schröder et al., 2023). The format can be used to guide the audience through a specific storyline, for example illustrating the gradual decline of infrastructure over time. Presenting data chronologically helps the audience understand what happened and why, reinforcing intervention urgency and consequences of delay (Kosara & Mackinlay, 2013). Additionally, the format allows contextual information, such as policy changes or funding decisions, to be presented alongside timelines, making their potential impact easier to interpret. Reports designed to support decision-making often emphasize overall quality, service levels, and budget impacts. Presenting this information clearly helps facilitate discussions around funding needs and investment sufficiency by linking infrastructure conditions directly to financial implications (Aijö et al., 2006). In the researcher's view, public-facing road condition reports can benefit from a linear narrative structure, for example by illustrating how a particular stretch of road has transitioned from 'good' to 'poor' condition over a decade, using time-series graphs or annotated maps to support the story. For example, in Finland, maintenance backlog indicators are communicated using time-series charts and annotated maps to clarify trends, targets, and policy options for both experts and the public (Hamalainen, Vaismaa, & Kolisoja, 2025). This approach helps make technical data accessible and supports informed decision-making.

The exploratory narrative format is defined by user-driven interaction, where audiences navigate data independently rather than following a fixed sequence (Kosara & Mackinlay, 2013). It often uses interactive visualizations and large-scale datasets more on the side of reader-driven storytelling. In the researcher's view, road engineers could benefit from the ability to interact with detailed maps, zooming into specific neighborhoods, or filtering distress type (e.g., potholes, cracks, rutting) to more precisely identify repair needs. Exploratory formats also align with cognitive principles of active learning and engagement (Schröder et al., 2023). To prevent overwhelming users with excessive options or a complex interface, this format benefits from thoughtful design elements such as filters, tooltips, and guided entry points that help balance freedom with usability (Alhamadi, 2020).

2.4 Process and Ethics of Storytelling

A significant part of data storytelling is about transparency of sources and explaining the analysis process (Oberascher et al., 2023). While visualizations may seem neutral, storytellers can unintentionally introduce personal viewpoints, for example, by choosing which aspects to emphasize, which can leave out important parts of the story (Schröder et al., 2023). To address this, storytellers should be aware of their biases and clearly explain data sources and analytical steps. Even small design choices like color, scale, or layout can subtly influence interpretation or reinforce stereotypes (Oberascher et al., 2023). An example of honest reporting is the ALARM report from the UK because it clearly shows where it gets its data, how it gathers it, and its limits (Asphalt Industry Alliance, 2025). Additionally, Ireland's trafficdata.tii.ie platform demonstrates openness by providing downloadable datasets and enabling users to independently verify and explore the information (Väyläviraston specialist, 2025; Transport Infrastructure Ireland, 2024). Oberascher et al. (2023) outline data storytelling as a five-step process for effectively showing and sharing open data. The first step, Discovery, involves exploring the dataset, reviewing metadata, checking for completeness, and identifying any outliers or anomalies, which helps in pinpointing key features for presentation. Next is Assistance, which focuses on the essential tasks of

cleaning and processing the raw data, preparing it for further analysis and presentation. The third step, Insight, is about creating visualizations, narratives, and social features to gather additional information from data owners and users, with the main goal of extracting and communicating insights effectively. Following this is Leverage, where conclusions drawn from the analysis are shared and discussed to estimate their validity and implications. Finally, the Trust step emphasizes the importance of openly sharing data sources and analytical methods to build credibility and prevent misinterpretation. This method not only supports open data use but also promotes honest data storytelling by ensuring that insights are clear and trustworthy.

2.5 Data Storytelling for Presentations

In presentations, each graph or fact should clearly answer a relevant audience question, with slide headlines stating the main takeaway to support quick understanding (Segel & Heer, 2010; Shao et al., 2024). Structuring the presentation around a logical narrative, for example, “Problem → Evidence → Impact → Solution,” helps audiences follow the story and understand the implications of the data (Kosara & Mackinlay, 2013). Building on this structure, connecting data to real-world experiences, such as demonstrating how poor road conditions cause longer delivery times, makes findings more relatable and memorable (Dykes, 2020).

Comparisons and benchmarks during presentations such as showing changes over time or regional differences, can help make the data easier for the audience to interpret (Segel & Heer, 2010). Visual tools like spider charts are particularly useful in a presentation setting for comparing performance against targets and illustrating gaps, especially when dealing with safety, budget, and accessibility (Ahonen et al., 2016). As shown in FIGURE 2, a bar chart can be used in a presentation to show a funding gap by comparing actual budgets to estimated maintenance needs over time.

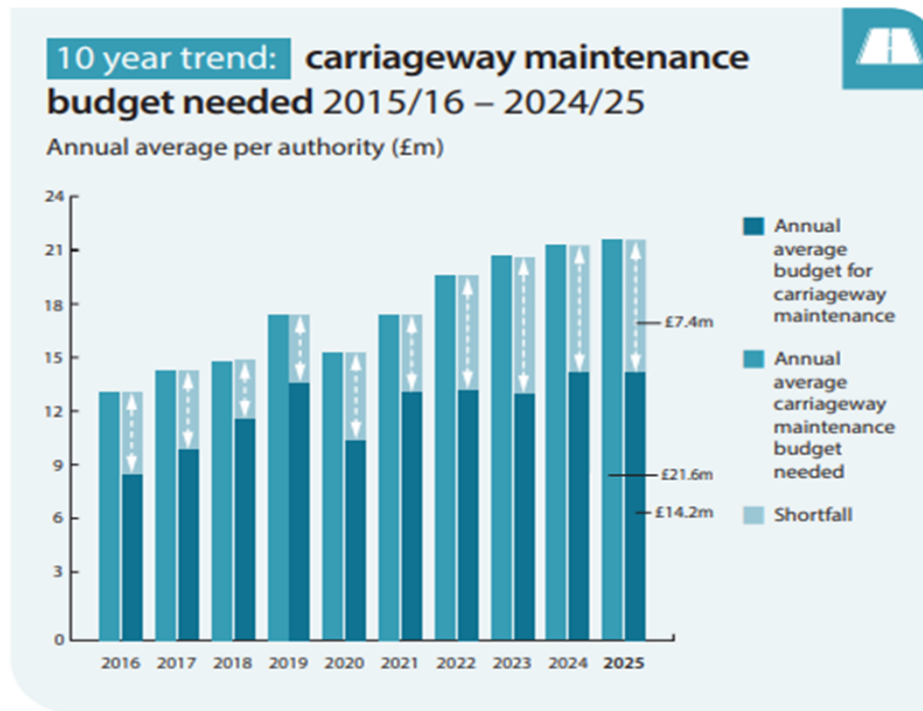


FIGURE 2. Annotated example slide: Funding Gap Analysis. Adapted from Annual Local Authority Road Maintenance (ALARM) Survey 2025, (Asphalt Industry Alliance, 2025).

2.5.1 Common Pitfalls in Data Storytelling

A common pitfall in data storytelling for presentations is overloading slides with excessive information or visuals, which can easily overwhelm the audience (Guo et al., 2018). In a presentation setting, where time and attention are limited, too much content on a slide can make it difficult for viewers to identify the main point (Guo et al., 2018). This often leads to cognitive overload, causing the audience to disengage. For example, a poorly designed slide might look like the image presented in FIGURE 3:

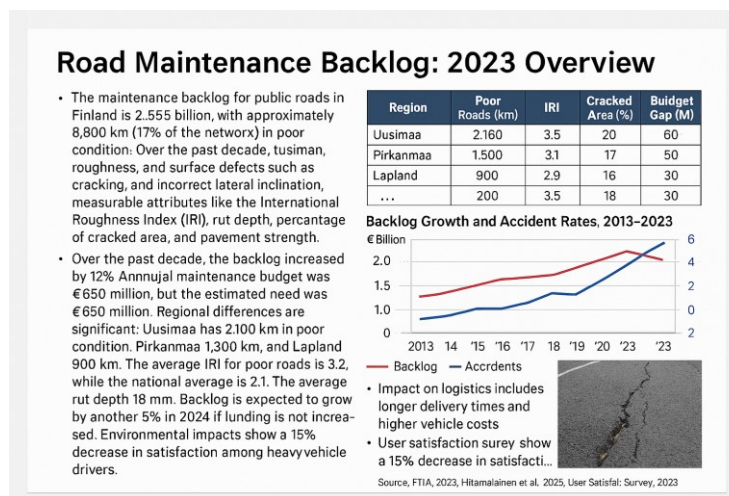


FIGURE 3. Created by the author (Suoninen, A., with illustrative data; generated using ChatGPT, 2025).

2.5.2 Accessibility in Presentations

In presentations, accessibility means ensuring that slides are usable and understandable for all stakeholders including those with visual or cognitive disabilities (W3C, 2023). This involves using high-contrast colors, readable fonts, and alt text to ensure visuals are accessible in different settings and for screen reader users (W3C, 2023). For instance, relying on only color to convey a message can exclude colorblind viewers. Additionally, small or decorative fonts may make it harder to read, especially from the back of the room.

Research shows that audiences in presentations benefit from clear guidance, such as suggested questions or step-by-step pathways that help them focus on the most important information (Segel & Heer, 2010). During presentations, presenters can actively engage the audience by asking open-ended questions, encouraging discussion, and inviting feedback on both the content and the visuals. Presenters can use audience responses to adjust their explanations and help build a sense of collaboration with the audience (Wood et al., 2014). This visual, FIGURE 4, created by the author with Microsoft Copilot, summarizes the author's own understanding of the main ideas for good data presentation: storytelling, slide design, accessibility, engaging the audience, and avoiding common mistakes:

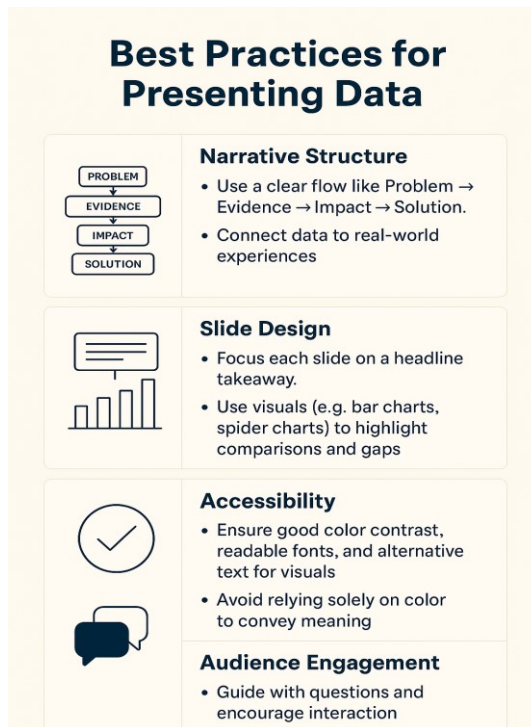


FIGURE 4. Created by the author (Suoninen, A., with illustrative data; generated using ChatGPT, 2025).

2.6 Collaborative Visualization and Evolving Stakeholders.

According to Wood et al. (2014), data visualization is a collaborative process involving multiple stakeholders, modes of interaction, and areas of application. Rather than using fixed step-by-step models, more flexible approaches that involve

a wider range of participants may be more effective (Wood et al., 2014). The concept of "multiple communication channels", also referred to as "channels of discourse", represents evolving relationships between those who create and those who consume visualizations. In this framework, multiple visualization projects occur at the same time for different users and purposes.

Wood et al. (2014), with a four-year engagement with London Cycle Hire Scheme (LCHS), identified different interrelated channels in which visualization played a central role. Each channel reflects distinct user needs and visualization goals. For example, within the transport operations channel, the primary goal was to achieve real-time monitoring to support immediate decision-making. This resulted in a gridded spatial dashboard displaying bicycle levels and 24-hour trends using sparklines.

In the policy and research channel, the objective was to facilitate deeper insight and strategic engagement among policymakers and transport researchers (Wood et al., 2014). To achieve this, a flexible visual analysis tool was developed to explore user behavior and spatial patterns. This tool was often used in a “chauffeuréd” setting, where researchers guided users through interactive visualizations. FIGURE 5 illustrates a mockup of such a dashboard, showing how coordinated views, filters, and tooltips can support guided exploration and decision-making.

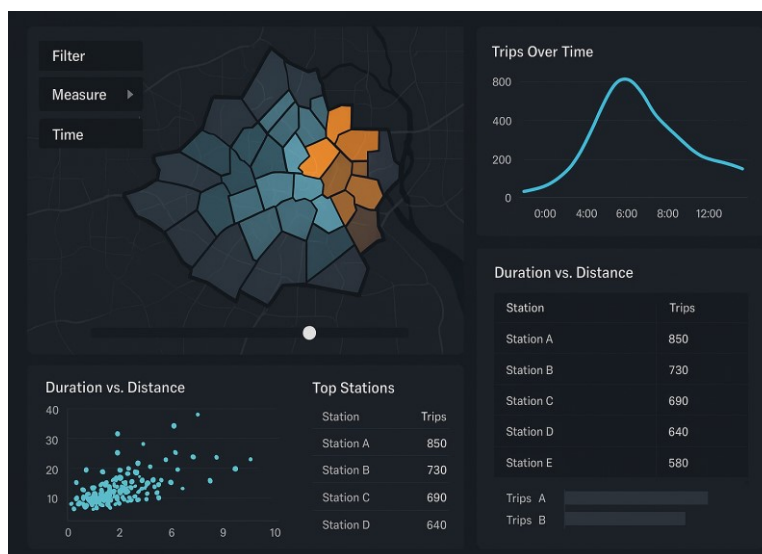


FIGURE 5. Created by the author (Suoninen, A., with illustrative data; generated using ChatGPT, 2025).

For public engagement, the focus was on accessibility and storytelling. An animated visualization of bicycle movements was developed using metaphors such as “light trails” to convey emergent urban structures from complex data (Wood et al., 2014). This output helped communicate the significance of big data and urban mobility to non-expert audiences. The animation became a “visualization meme” within the organization, also adopted by policymakers to explain data use and strategy (Wood et al., 2014).

This project shows how distinct visualization outputs can come from the same dataset and serve different functions. The efforts to visualize the data were interconnected, informed, and reinforced by one another. Wood et al. (2014) suggest that using multiple channels improves visualization by allowing designers to customize outputs for different audiences while sharing data and insights.

2.6.1 Design-Based Research Approach

Kempeneer et al. (2025) presents a framework for improving citizen understanding of Open Government Data (OGD), utilizing a design experiment in the form of a field lab to foster innovation in a real-life context. This approach includes a diverse group of participants: employees from a regional water authority (acting as government data providers and consumers), data visualization experts (civil programmers), and non-expert citizens, and researchers (Kempeneer et al., 2025). The inclusion of non-expert citizens is particularly emphasized as they are the target audience for new forms of OGD dissemination. The methodological framework is structured according to Hevner's (2007) three-cycle design science model:

Relevance Cycle (Phases 1-3): Identifying the problem of OGD dissemination to non-experts, exploring the context, stakeholders, and their needs and challenges. In the field lab, this involved observing the target group, exploring problem definitions related to drought, discussing current communication forms, and brainstorming alternative communication strategies.

Design Cycle (Phase 4): Data experts create a prototype solution, to improve OGD dissemination. This phase can be repeated if feedback requires further design iterations.

Rigour Cycle (Phase 5): Evaluating the designed prototype with field lab participants along specific dimensions. In Kempeneer's study, the evaluation focused on completeness, objectivity, and usability, often through a final field lab session where government employees and citizens provided feedback on the prototype.

This framework highlights the importance of user design, understanding what stakeholders need, and the idea of constantly gathering feedback and making improvements to stay relevant over time.

2.7 Chatbots in the Public Sector: Microsoft Copilot Studio

Public sector organizations are increasingly adopting chatbots to deliver information and services, reflecting a broader shift toward digital government and the growing digitalization of service provision (Abbas, Følstad, & Bjørkli, 2023). Digital government initiatives aim to reduce costs and improve the quality of information and services provided to citizens, with the hope of increasing user satisfaction. (Abbas et al., 2023).

AI-powered chatbots use natural language processing (NLP) to simulate human-like conversations, enabling them to understand, analyze, and respond in ways that feel intuitive to users (Hosseini & Tupasela, 2024). Their primary goal is to make public data more discoverable and understandable through plain language, intuitive visuals, and data storytelling techniques (Microsoft Corporation, 2024). These capabilities position chatbots as key tools for improving accessibility and transparency in public sector communication.

2.7.1 Microsoft Copilot Studio and the Agent Paradigm

Microsoft Copilot Studio applies Generative Pre-trained Transformer (GPT) based models to build virtual agents that function more dynamically than fixed

chatbot scripts. (Riley, 2024; Microsoft, 2025). Microsoft refers to these as “agents” to emphasize their ability to operate more independently and intelligently (Riley, 2024). These agents coordinate multiple Artificial Intelligence (AI) components such as language models, contextual information, knowledge sources, and triggers to achieve specific goals (Microsoft, 2025). A key feature of Copilot agents is their ability to “network” across public sector databases (Kaun & Männiste, 2025). For example, when a user asks a local government chatbot about current road conditions, the system can combine local data with information from national infrastructure or meteorological databases. This interoperability helps the agent give clear, correct, and relevant answers, improving the quality of public service delivery (Kaun & Männiste, 2025).

2.7.2 Personalization and Learning Support

Beyond system integration, Copilot agents enable personalized guidance through user profiling and adaptive learning. By analyzing previous interactions, agents can assess a user’s initial knowledge level and tailor responses accordingly (Hosseini & Tupasela, 2024). This approach allows the system to deliver targeted advice and resources that match the user’s background and current context. For instance, a chatbot that remembers past inquiries can improve future interactions by recalling stored conversation records (Hosseini & Tupasela, 2024).

To further enhance personalization, agents can employ pre-assessments to figure out a user’s starting point and adjust content complexity (Hosseini & Tupasela, 2024). If a new staff member does not know some of the basic terms, the system can help them by giving simple step-by-step instructions for complicated tasks. This method, known as scaffolding, bridges new information with existing knowledge, supporting gradual skill development, and improving digital literacy over time (Hosseini & Tupasela, 2024).

2.7.3 Data Visualization and Preprocessing with Copilot Agents

In addition to guiding users step by step, Copilot agents can convert everyday questions into visualizations like charts and graphs (Chokki, 2023). The system

first analyses data types: numeric, categorical, temporal, or geographic. Then it can create heatmaps, word clouds, and correlograms to help users understand data relationships (Microsoft, 2025). It supports many chart types, including bar, line, sunburst, and parallel coordinates, and can create dashboards with multiple visuals (Alhamadi, 2020). Before visualization, users can preprocess data by dropping columns, merging datasets, or transforming values (Chokki et al., 2022). Copilot also creates basic narratives for each visualization that users can edit to match their needs and interpretations (Chokki et al., 2022).

2.7.4 Chatbot Feedback Loops and Continuous Improvement

Copilot Studio's chatbot allows users to give feedback like ratings or comments on suggested visualizations, helping them improve future recommendations (Chokki et al., 2022). Such feedback loops are critical for identifying and addressing any barriers to accessibility or inclusiveness and assisting continuous enhancement of the chatbot's functionality (Hosseini & Tupasela, 2024).

A practical application of this improvement process is seen in the City of Helsinki, which uses a practice of using "chatbot whisperers" (Hosseini & Tupasela, 2024). The system records the questions asked and the answers given during each session, and this log is systematically analyzed by the service development team (Hosseini & Tupasela, 2024). This analysis helps identify areas where the training material and the chatbot's answers need improvement (Hosseini & Tupasela, 2024). Based on this analysis and customer feedback, the training material is refined, more sample questions can be added, or new discussion paths can be created to improve the user experience (Hosseini & Tupasela, 2024).

This approach reflects the broader principles of user-oriented reporting and design-based research emphasized throughout this thesis. By actively incorporating stakeholder feedback and analyzing real-world usage, organizations can ensure that digital tools like chatbots remain accessible, inclusive, and responsive to evolving user needs. Continuous improvement, driven by open feedback channels and systematic evaluation, is critical for maintaining trust and relevance in public sector communication, whether in chatbot interfaces or infrastructure reporting templates.

2.8 Data Storytelling Process: Exploration, Scripting, Delivery

Schröder et al. (2023) describe a three-stage framework for effective data storytelling: data exploration, where analysts uncover key trends; narrative scripting, where communicators make storylines and emphasize relevant insights; and story delivery, where messages are designed for a specific audience and chosen medium. This framework highlights the collaborative and iterative nature of storytelling with data, emphasizing that meaningful communication requires more than just presenting numbers.

In the data exploration phase, analysts examine how raw datasets show patterns, anomalies, and correlations that are both statistically and contextually significant (Schröder et al., 2023). This step also involves considering what matters most to the audience. As noted by Hamalainen et al. (2025), stakeholder feedback, including citizen complaints, can provide valuable insights for transportation planning that may not be evident from quantitative traffic data.

The narrative scripting phase involves selecting and organizing insights into a coherent storyline. Specialists must consider the audience's background, interests, and informational needs to ensure the information is relevant to the audience. By framing data in ways that connect with real-world concerns such as safety, accessibility, or environmental impact, the narrative becomes more persuasive (Schröder et al., 2023). The researcher would suggest that connecting infrastructure data to real-world outcomes, such as emergency response delays, can make reports more compelling and actionable.

In the story delivery phase, the narrative is adapted to the chosen medium, whether it be a dashboard, report, presentation, or social media post (Schröder et al., 2023). Effective delivery involves thoughtful design choices, including the use of visual hierarchy, color, and annotation, to ensure clarity and impact. Then tailoring the message to the audience, such as using simple visuals for the public and detailed charts for policymakers, makes the message easier to understand and remember (Schröder et al., 2023). For example, bright colors on maps can quickly show road conditions, and visual cues like arrows, bold lines, or contrast

help highlight key trends. Titles and annotations also play a crucial role: instead of a generic heading like “Road Condition Map”, descriptive titles such as “Caution: Major Roads Are in Poor Condition” or “The Pothole Problem on Hämeenkatu” immediately communicate the message. However, the amount of text must be carefully balanced with visuals, as excessive annotation can confuse or distract viewers (Shao et al., 2024).

2.9 Infrastructure Asset Management (ISO 55001)

The International Standard ISO/FDIS 55001:2024 on Asset Management provides a framework for establishing, operating, and improving asset management systems (SFS Finnish Standards, 2025, pp. 5, 28). In the context of this study, the most relevant aspects for Väylävirasto concern communication and stakeholder engagement. The standard calls for organizations to define what, when, with whom, and how to communicate, and to regularly evaluate whether communication meets stakeholder needs (SFS Finnish Standards, 2025, Clause 7.4, pp. 18, 40). It also emphasizes identifying stakeholders’ “needs and expectations” and considering the impacts of asset management activities in decision-making, prioritization, and performance monitoring (SFS Finnish Standards, 2025, Clause 4.2, pp. 11, 33). These provisions support Väylävirasto in tailoring its reporting and data storytelling to diverse audiences with varying levels of data literacy.

ISO 55000 is not a mandatory requirement for Finnish public-sector organizations, and its adoption can vary between organizations. Väylävirasto has selectively applied parts of the standard. Instead of pursuing full certification, the agency incorporates key principles such as structured documentation and effective practices in communication, which allow it to benefit from international guidance while maintaining flexibility (Väylävirasto specialist, 2025).

A key part of public reporting is making sure that information and terminology are consistent and connected between the financial and non-financial parts of the organization (SFS Finnish Standards, 2025, Clause 7.6, p. 19). In cases where differences are necessary, this information must be made available as documented information (SFS Finnish Standards, 2025, Clause 7.6, p. 19). To achieve

this, organizations must define and establish specifications for information, covering attributes, units of measurement, quality, and source, and develop a plan for sharing the data and information (SFS Finnish Standards, 2025, Clause 7.6, p. 19, p. 41).

ISO 55001 mandates version control, review and approval workflows, controlled access, and reliable distribution (SFS Finnish Standards, 2025, Clauses 7.5.2–7.5.3, pp. 40–41). These controls reinforce the standard's goal of "confidence in information," (SFS Finnish Standards, 2025, Clause 4.5.3, p. 34). Confidence depends on systematic handling of both communication processes and information quality (SFS Finnish Standards, 2025, Clauses 7.4 & 7.6, pp. 40–41), which is key to aligning operational data with stakeholder needs.

Beyond communication and documentation, ISO 55001 emphasizes the importance of ongoing performance evaluation, for example internal audits, and continual improvement (SFS Finnish Standards, 2025, Clause 10.1, p. 45). Clause 9.1 requires organizations to monitor and evaluate performance not just technically, but also in terms of stakeholder needs and the effectiveness of communications (SFS Finnish Standards, 2025, Clause 9.1, p. 43; Clause 7.4, p. 40).

Finally, Clause 9.3 mandates regular assessments of changing stakeholder expectations (SFS Finnish Standards, 2025, Clause 9.3.2, p. 44), evaluation of asset management performance (SFS Finnish Standards, 2025, Clause 9.3.2, p. 44), and identification of areas for improvement (SFS Finnish Standards, 2025, Clause 9.3.2, p. 44). ISO 55001 describes communication as a repeating; ongoing process that's key to making sure reports on public infrastructure stay relevant and trustworthy over time.

2.9.1 Legal Frameworks and Standards

The EU Accessibility Act (Directive (EU) 2019/882) sets legal requirements to ensure digital products and services are accessible to all users, especially people with disabilities (European Union, 2019). This includes websites, digital documents, and online services. As a result, public infrastructure reports must use

clear language, accessible visuals, and formats that work with assistive technologies (Laamanen et al., 2022). The Web Content Accessibility Guidelines (WCAG) further specifies criteria to ensure accessibility for users with diverse abilities and levels of technical skill, including areas such as color contrast, alternative text for graphics, keyboard navigation, and readable typography (W3C, 2023).

The European Union's INSPIRE Directive (2007/2/EC), established in 2007, has driven EU members to organize their spatial data using common rules, standardized data models, and open formats. Standardized data structures support international collaboration in infrastructure planning by enabling cross-border data sharing and informed policymaking (Shevtsova, 2023). For instance, when countries use similar formats for road data, they can more easily compare and integrate information

Finland has incorporated these EU-level directives into national law through the Act on the Provision of Digital Services (306/2019), which mandates public-sector organizations and relevant private entities to ensure that their digital services meet accessibility requirements. An important technical standard of compliance is EN 301 549, which integrates the WCAG 2.1 guidelines. These guidelines use four foundational principles: perceivable, operable, understandable, and robust. In practice, this means that content must be perceivable by providing alternatives for non-text elements, distinguishable design features; operable through full keyboard accessibility, understandable with clear text, and input support; and updatable that it functions across devices and assistive technologies for long-term compatibility (Kempeneer, Janssen, & Schmitz, 2022).

2.10 International Practices in Infrastructure Reporting

Across Europe, infrastructure condition reports such as the UK's ALARM report and Ireland's TII Asset Management Plan demonstrate how complex technical data can be made accessible to diverse stakeholders through clear visual and structural design (Asphalt Industry Alliance, 2025; Transport Infrastructure Ireland, 2023). While benchmarking is challenging due to varying asset types and mandates (Konstantakos, Chountalas, & Magoutas, 2019), these reports among

others demonstrate how visual layouts, dashboards, and interactive formats can make data more accessible to non-technical audiences, supporting transparency and engagement (Kempeneer et al., 2022).

Both ALARM report and TII reports distill technical information into modular, audience-friendly formats. For instance, ALARM's one-topic-per-page approach and TII's Key Trends Summary use concise headlines, targeted statistics, and visual cues such as traffic light colors, icons, and arrows to highlight trends and priorities (Asphalt Industry Alliance, 2025; Transport Infrastructure Ireland, 2023). A typical example is ALARM's "Road Condition Index" visualization, which presents the proportion of roads in good, fair, or poor conditions across England, Wales, and London, comparing actual performance to targets (see FIGURE 6).

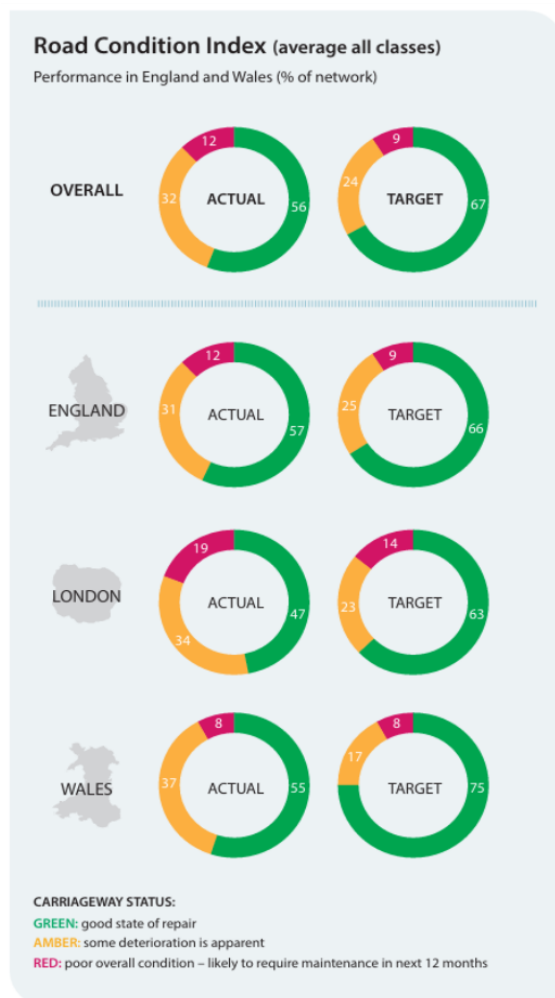


FIGURE 6. Road Condition Index (average all classes) Performance in England and Wales (% of network). Adapted from Annual Local Authority Road Maintenance (ALARM) Survey 2025 (Asphalt Industry Alliance, 2025).

The ALARM report uses traffic light colors, icons with brief labels, and arrows to indicate condition severity and trend direction (Asphalt Industry Alliance, 2025). This is shown in the "Key findings" summary table on page 24. For example, arrows denote whether the "Average highway maintenance budget per authority" has gone down, or the "Shortfall in road carriageway budget" has gone up.

Key findings				
	TOTAL*	England**	London	Wales
Percentage of authorities responding	↑ 78%	↑ 83%	↑ 78%	→ 50%
Highway maintenance budgets				
Average highway maintenance budget per authority	↓ £26.0m	↓ £33.5m	↑ £10.8m	↑ £8.5m
Percentage of highway maintenance budget spent on carriageway	↑ 53%	↑ 56%	↑ 42%	↓ 54%
Average carriageway maintenance budget per authority	↑ £14.2m	→ £18.7m	↑ £4.5m	↑ £4.6m

FIGURE 7. Key findings summary showing average highway maintenance budgets and spending by region. Adapted from Annual Local Authority Road Maintenance (ALARM) Survey 2025 (Asphalt Industry Alliance, 2025).

Similarly, in Finland, concise visuals and headline indicators are used to communicate the road maintenance backlog, supporting stakeholder engagement and transparency (Hamalainen et al., 2025). These practices reinforce the importance of clarity and accessibility in infrastructure reporting.

Using these international practices makes reports clearer and more useful for all users. By learning from these examples, organizations like Väylävirasto can make their reports easier to understand and more relevant for both experts and the general public.

2.11 Synthesis of Theoretical Framework

To bring together the main ideas from the literature, FIGURE 8, below summarizes the key theoretical elements of user-oriented infrastructure reporting which

is the key output of this thesis. The figure shows how data storytelling, visualization practices, chatbots, standards, international practices, and asset management are all connected in this field.



FIGURE 8. Concept map of key theoretical elements. Created by the author using Miro, October 2025.

This overview helps clarify the main concepts and shows how they relate to each other. It also provides a clear foundation for the next sections, where the research methods and approach will be explained.

2.11.1 Annotated Visuals and Narrative Headlines

A key point that came up in the literature is how effective it is to use annotated visuals together with narrative headlines in infrastructure reports. Instead of using generic titles like “Road Condition Map,” use headlines such as “Caution: Major Roads Are in Poor Condition,” which helps readers immediately understand the main message (Asphalt Industry Alliance, 2025; Schröder et al., 2023). Additionally, adding annotations like arrows, callouts, or short notes directly onto charts or maps makes it easier to spot important trends or unusual results.

3 METHODOLOGY

In this section, the overall methodology used for the thesis is introduced. The focus is on explaining the reasoning behind the chosen research approach and outlining how different methods, like design-based research, stakeholder engagement, and benchmarking, were combined to support the investigation. The aim is to give a clear overview of how the research was structured to address the main question: how can infrastructure reports be made more user-oriented and accessible for a range of audiences?

3.1 Case Study Design

This research adopts a case study approach to examine how Vöylävirasto can improve its public communication of road and bridge condition data. The project was conducted in a consulting-style collaboration with Vöylävirasto, fulfilling two main roles: (1) co-designing and iteratively prototyping a stakeholder-oriented report template (2) documenting the design process and outcomes within a thesis, supported by academic literature and established methodological frameworks.

The study follows an iterative design-based research (DBR) framework, drawing on stakeholder-oriented co-design, data storytelling, and benchmarking against international good practices to develop and refine new reporting elements and practices (Barab & Squire, 2004; Hevner, 2007). DBR was chosen because it enables the integration of theory and practice in real-world settings, allowing iterative cycles of design, implementation, and evaluation. Unlike traditional experimental approaches, DBR supports collaborative problem-solving with stakeholders and adapts to evolving requirements, making it particularly suitable for addressing complex communication challenges in infrastructure reporting.

In addition to iterative co-design, the methodology was informed by literature on narrative framing and audience-specific visualization. Throughout, the evolving design elements and practices were evaluated against international standards to ensure practical relevance and methodological rigor. Ethical guidelines were strictly followed, with informed consent obtained from all participants and secure

handling of data throughout the research. The research also prioritized accessibility, minimization of visual bias, and transparency in reporting, aligning design choices with ISO 55001 and the Web Content Accessibility Guidelines to ensure equitable access and trustworthy communication. The research focuses on Vöylävirasto's reporting activities from March 2025 to November 2025. Vöylävirasto was selected because it manages and shares large amounts of technical infrastructure data with diverse stakeholders, making it an appropriate setting to explore methods for improving the accessibility of technical information.

Stakeholder groups were identified through a systematic process combining qualitative interviews, internal document analysis, and supervisor recommendations. The research began with a systematic assessment of stakeholder needs and communication challenges, identified through semi-structured interviews with staff involved in asset management, communications, and data visualization. Internal documents and previous reports were reviewed to map existing practices and highlight information gaps. Data collection ended after nine interviews with Vöylävirasto staff, as no new important themes were emerging, suggesting that saturation was reached.

3.2 International Benchmarking

As stated, insights from international benchmarks, including the UK's ALARM report and Ireland's Strategic Asset Management Plan, were used to guide design strategies and ensure solutions were practical and based on effective practices. These benchmarks were chosen for their ability to present complex infrastructure information clearly to non-technical audiences. The UK's ALARM report and Ireland's Asset Management Pla are frequently referenced in the literature for their clear visuals, modular structure, and audience-friendly design (Asphalt Industry Alliance, 2025; Transport Infrastructure Ireland, 2023; Kempeneer et al., 2022). Although direct, published evaluations of stakeholder feedback on these reports are not available, their adoption and citation as effective examples in both academic and professional contexts suggest strong acceptance and influence on similar frameworks (Kempeneer et al., 2022; Schröder et al., 2023).

3.2.1 Hevner's Three-Cycle Model

An important methodological framework for this case study is structured according to Hevner's (2007) Three-Cycle View of Design Science Research, encompassing the Relevance Cycle, the Design Cycle, and the Rigor Cycle. The main research activities for this study are mapped in TABLE 1 below, organized according to these three cycles.

TABLE 1. Mapping of research activities to design science cycles. Adapted from Hevner (2007) and developed by the author.

Cycle	Research Activities
Relevance	Stakeholder interviews, internal document analysis
Design	Prototype development, iterative feedback loops
Rigor	Benchmarking, literature integration, usability checks

Specifically, the Relevance Cycle involved stakeholder interviews and internal document analysis to identify communication challenges and stakeholder needs; the Design Cycle comprised prototype development and iterative feedback loops with Vöylävirasto staff; and the Rigor Cycle included benchmarking against international reports, integration of relevant literature, and usability checks to ensure the robustness and adaptability of the report template (Hevner, 2007). The report template was developed through three design iterations, following the structure of Hevner's Three-Cycle View. After each iteration, feedback was systematically gathered from Vöylävirasto staff, focusing on language, visual formats, and content structure, and iterative adjustments were made based on stakeholder responses. The specific outcomes of these iterations, including concrete template changes and stakeholder reactions, are detailed in the Findings section.

3.3 Stakeholder Engagement and Evaluation

The Relevance Cycle, as outlined in Hevner's design science framework, connects the research context to design science activities by systematically identifying the challenges of communicating complex road and bridge data to non-expert citizens, mapping the specific context, key stakeholders, their diverse needs, and existing barriers (Hevner, 2007). The development process involved continuously building and refining the prototype, with each iteration evaluated against evolving requirements and informed by stakeholder feedback. Interview transcripts were analyzed through manual thematic coding so that patterns and themes could arise directly from the data. This approach supported the identification of recurring themes related to stakeholder needs, communication barriers, and design preferences, which directly informed the iterative refinement of the report template. Once the main stakeholder needs had been established, the reporting template was developed with a strong focus on making it genuinely useful for both internal and external audiences. Feedback from Vöylävirasto staff highlighted the importance of a report template that would help users outside the organization get involved, ask questions, and share information more easily. This input guided the design work throughout the process.

3.3.1 Evaluating the Report Template

When evaluating the report template, attention was given to how complete, easy to use, and objective it was, following public sector reporting standards (Hevner, 2007). Regular management reviews and opportunities for improvement, as outlined in Suomen Standardit ry (SFS) Finnish Standards (2025, Clause 9.3), were integrated throughout the process to support continual refinement and ensure the reporting prototype's relevance and comprehensibility over time. Asset specialists offered specific advice about headlines, layout, and how well the slide was about to convey a story through the data and narrative. For more details about what was changed and how stakeholders responded, see the Findings section.

3.4 Potential of Chatbots

While the literature review explores the potential of Chatbots and Copilot Studio for public sector communication, these technologies were not implemented in the case study or report template. Microsoft Copilot, created by the author, was used only to assist with the creation of presentation visuals, such as brainstorming visualizations and generating tables. All Copilot-generated content was produced under the author's supervision to ensure alignment with research objectives. Its use was limited to supporting visual design, not interactive or automated communication. Future research could explore the integration of generative artificial intelligence or Copilot agents for citizen feedback or Frequently Asked Questions (FAQ) triage, as discussed in the theoretical framework.

Prototype visuals were created and shared with stakeholders to gauge clarity and usefulness. One example, shown in FIGURE 9, is a scatterplot illustrating the relationship between maintenance backlog growth and traffic volume across regions. This scatterplot was generated with the assistance of Microsoft Copilot, which helped automate data visualization and formatting tasks under the author's guidance. Scatterplots were chosen for their ability to reveal patterns and outliers, allowing regional comparisons immediately (Wood et al., 2014).

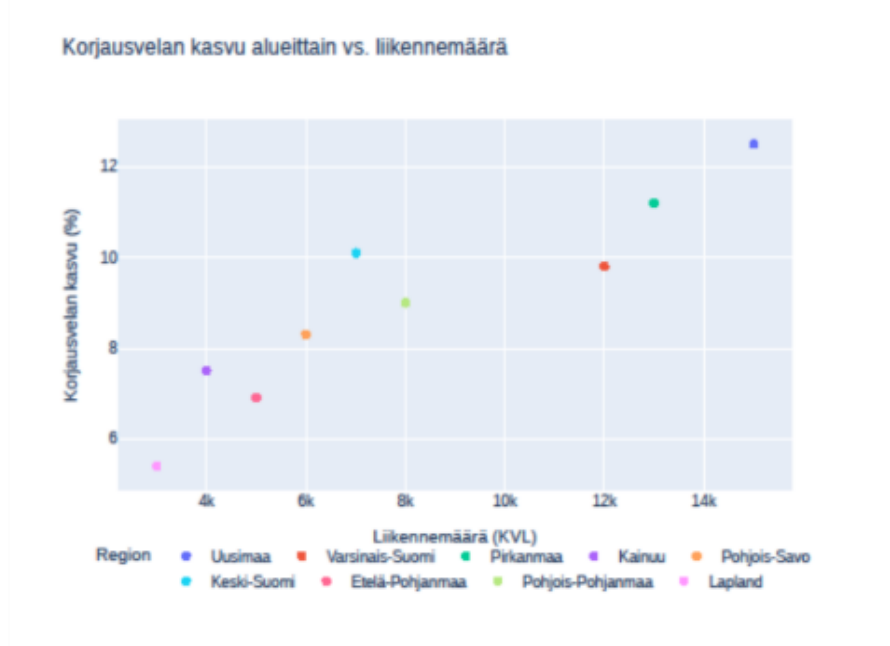


FIGURE 9. A prototype map used in stakeholder sessions to test the effectiveness of multi-criteria visual summaries. Created by the author.

3.4.1 Practical and Conceptual Contributions

This study was designed to generate both practical and conceptual insights into stakeholder-oriented infrastructure reporting. The practical objective was to develop a reusable report template and a set of design guidelines for Väylävirasto, supporting effective communication with diverse stakeholder groups, including policymakers, citizens, and journalists, each of whom has distinct information needs. Conceptually, the research investigates how data storytelling methods, informed by qualitative interviews, international benchmarking, and iterative prototyping, can enhance public understanding of technical infrastructure data. The methodological approach integrates design-based research principles and case study analysis to ensure that the report template and guidelines are grounded in stakeholder feedback and best practices. Evidence-based recommendations for user-centered and iterative communication practices developed through this process are presented in the Findings section.

3.4.2 Selection of Stakeholder Interviewees

Interviewees were chosen using purposeful sampling to cover roles most relevant to reporting and visualization, ensuring perspectives from asset management, communications, and data analysis were included. Interviewees were chosen based on staff identification and supervisor recommendations to include those with experience in reporting, visualization, and stakeholder engagement. This ensured interviews reflected varied perspectives on current practices and challenges in infrastructure data communication.

3.4.3 Data Collection: Qualitative Interviews, Case Analysis, and Feedback Cycles

Data collection for this study followed a qualitative, design-based research approach. Primary data were collected through semi-structured interviews with

Väylävirasto staff involved in asset management, communications, and data visualization. Interviews were conducted via video calls and email between May and November 2025.

Secondary data were obtained through systematic benchmarking of internationally recognized infrastructure reports, specifically the UK's ALARM report and Ireland's Strategic Asset Management Plan (SAMP). These documents were reviewed for their layout, use of visuals, and narrative framing, with the aim of identifying design strategies applicable to the Finnish context. In addition, common stakeholder questions and previous agency presentations were analyzed to inform the report template's structure and content, ensuring that the final design addressed both recurring information needs and communication challenges.

The interview protocol was designed to gather detailed information about current reporting practices, communication barriers, and stakeholder needs. Thematic analysis was applied to interview transcripts and benchmarking notes to find recurring patterns and refine the report template slide by slide. For a detailed account of specific findings and the influence of these insights on the template development, see the Findings section.

3.5 Content Decisions

The report template was developed based on a review of current reporting practices and input from staff who work with stakeholders. Conversations were held with customer service, communications, and technical teams to learn what questions they get most often and what information stakeholders need, especially policy decisions. These insights shaped the topics, highlighting areas like economic impacts, accessibility, and regional equity that were missing in previous reports. Finally, the content was refined to match what the senior specialist director found most relevant and engaging.

Design choices prioritized simplicity and clarity, using single-topic slides and scorecard-style summaries for easy interpretation. The report template includes a 'Key Insights' section, inspired by international practices such as Ireland's Stra-

tegic Asset Management Plan. This section highlights the most important findings, so policymakers can quickly grasp critical information. By presenting essential messages concisely and prominently, it applies audience-focused communication principles and adapts proven global strategies to the Finnish context.

3.5.1 Qualitative Interview Design and Justification

Semi-structured interviews were used to explore how technical infrastructure information could be made more understandable for different stakeholder groups. This approach was chosen because it allows for open-ended questions and follow-up discussion, making it possible to gather both expected and unexpected insights into communication needs and challenges.

The interview questions were designed to match the main research objectives and to reflect the theoretical framework, especially ideas about data storytelling and plain language. For example, participants were asked about their roles, the types of information they need, and what they find difficult or confusing in current reports. Questions about the use of storytelling and visuals were also included, since these are central themes in the thesis. Some interview questions were also inspired by the five-step data storytelling process described by Oberascher et al. (2023), which includes Discovery, Assistance, Insight, Leverage, and Trust. This framework helped shape questions about how data is explored, prepared, communicated, and how trust is built with different audiences.

Semi-structured interviews were selected because they are well-suited for exploring complex topics and allow for flexibility in the conversation (Hamalainen, 2025). Both video calls and email interviews were used, depending on what was most convenient for each participant. Email interviews gave participants more time to reflect on their answers, which sometimes led to more detailed responses.

As the research progressed, the interview questions were refined based on what was learned from earlier interviews. If certain topics or challenges came up repeatedly, these were explored further in later interviews. This ongoing adjustment helped keep the interviews relevant and ensured that the data collected would be useful for the next steps in developing the reporting template.

3.6 International Benchmarks in Infrastructure Communication

To evaluate the practical application of theories concerning visual clarity, stakeholder alignment, and narrative structure, the researcher conducted international benchmarking, focusing on established public infrastructure reports such as the UK's ALARM report and Ireland's Strategic Asset Management Plan (SAMP), both widely recognized for their effective communication with diverse audiences (Asphalt Industry Alliance, 2025; Transport Infrastructure Ireland, 2023). Each report was systematically reviewed, with detailed notes taken on the organization of information; the visual formats employed including dashboards, maps, charts, and scorecards, and the methods used to foreground key messages. Particular attention was devoted to features that enhanced clarity, accessibility, and narrative framing for both expert and non-expert audiences.

The benchmarking process was designed not to transfer content directly, but to identify and critically evaluate visual and narrative techniques that could be adapted to the Finnish context. To ensure contextual relevance, the strategies identified in these international reports were compared with the needs and preferences expressed by Väylävirasto stakeholders during semi-structured interviews. This comparative analysis informed the selection of approaches considered suitable for integration into the stakeholder-oriented report template. The specific elements adapted from these benchmarks, and their impact on the final report template design, are detailed in the Findings section.

3.6.1 Ethical Considerations

All participants provided informed consent prior to their interviews, either in writing or verbally, after being fully briefed on the study's purpose and the intended use of their contributions. The research did not involve the collection of personal or sensitive data; instead, it focused on public information, such as road condition statistics and infrastructure metrics. Interview notes and recordings were securely

stored on the university's encrypted server, in compliance with General Data Protection Regulation (GDPR) and the research ethics guidelines of TAMK and the Finnish National Board on Research Integrity (TENK, 2023). Any quotations included in the thesis were anonymized to protect participant identities. For benchmarking activities, only publicly available reports were consulted, ensuring adherence to copyright and data protection requirements.

The research process was conducted in accordance with the principles of Responsible Conduct of Research (RCR) as outlined by TENK (2023), emphasizing integrity, meticulousness, and accuracy throughout data acquisition, analysis, and reporting. The aim was to provide an honest and constructive case study of communication challenges and opportunities in public infrastructure reporting, with identified issues framed as areas for development rather than criticism. Research results were communicated openly and responsibly, consistent with established scientific dissemination practices. As the thesis is a public document, care was taken to ensure that no confidential or classified information was included in the final report. In accordance with the thesis contract, confidential information from the cooperation partner or protected by law was not disclosed, with this obligation extending for five years from the contract's commencement (Finnish National Board on Research Integrity [TENK], 2023). Ethical considerations were integrated throughout the iterative prototyping and feedback cycles, with participant input handled confidentially at each stage of the design process.

3.7 Data Analysis: Thematic Coding

Interview transcripts were analyzed using manual thematic coding, with codes and categories developed inductively from the content of participant responses. Given the limited number of interviews, a lightweight manual approach using Microsoft Excel was adopted to systematically organize, group, and compare the data. The thematic analysis concentrated on identifying recurring patterns related to visual clarity, stakeholder needs, communication barriers, and organizational tools or capacity.

TABLE 2. Summarized Main Themes Identified in Thematic Analysis and Their Influence on Report Template Design

Theme	Influence on Report Template Design
Visual Clarity	Use of color-coded charts, simple layouts
Stakeholder Needs	Modular FAQ structure, plain language
Communication Barriers	Glossary, annotated visuals, clear headlines
Tools/Capacity	Editable PowerPoint format, reusable slides

To enhance the robustness of the analysis, insights from international benchmarking were systematically compared with the emergent themes, allowing for the identification of both overlaps and gaps between Väylävirasto's current practices and recognized effective practices abroad. This integrated approach ensured that the combined findings from interviews and benchmarking directly informed the iterative design and refinement of the stakeholder-oriented report template. To further ensure transparency and methodological rigor, coding decisions and emergent themes were regularly reviewed and discussed with project collaborators, supporting reflexivity and minimizing potential bias. The integration of interview and benchmarking insights was embedded within the project's iterative feedback cycles, allowing for continuous refinement of both the analytic framework and the report template itself. The specific themes identified, and their influence on the report template structure and content, are discussed in detail in the Findings section.

3.7.1 Development Process

An internal document analysis was carried out to gain insight into current reporting practices, technical definitions, and data structures related to pavement and

bridge condition monitoring in Finland. The primary sources reviewed included “Päällystettyjen teiden” FTIA Instructions 10/2021, which describe policies and measurement protocols for paved roads, and “VÄYLÄVIRASTON SILLAT” FTIA Publications 67/2024, which outline bridge condition metrics and recent updates to classification systems. These documents provided the technical foundations that were incorporated into the Condition Status, Technical Indicators, and Bridge Condition sections of the stakeholder-oriented report template.

In addition to these official instructions, internal presentations and historical technical references were consulted to ensure that the report template reflected both current standards and the development of infrastructure management practices over time. This approach helped to avoid the use of outdated information and supported the accuracy of the report template.

To strengthen the analysis, findings from the document review were triangulated with qualitative interview data and international benchmarking. Through this process, recurring patterns and gaps in existing reporting practices were identified and addressed during iterative prototyping and stakeholder feedback cycles. Technical definitions and measurement protocols from Vaylavorasto’s documents were translated into accessible language and visual formats within the report template, as shown in slides 14–17. This method is consistent with established practices in design-based research, where document analysis is used to ground artefact development in organizational context and technical standards (Bowen, 2009; Barab & Squire, 2004).

3.7.2 Historical Technical References

To better understand how pavement and bridge management practices developed in Finland, historical technical references were reviewed, such as Infrastructure Management System: Case Study of the Finnish National Road Administration (Mannisto & Tapio, 1994) and the Finnish Bridge Management System (Söderqvist & Veijola, 1991). These documents helped clarify how key technical indicators like pavement roughness, rutting, surface defects, and bridge condition attributes are defined and measured. By grounding the work in these established definitions, it was ensured that the reporting template accurately reflected both

current standards and the evolution of infrastructure management in Finland. Technical concepts were then translated into accessible language and visuals within the report template, making them easier for non-technical audiences to understand. Additional details on how these indicators are communicated to stakeholders are provided in the Findings section.

3.7.3 Incorporation of Internal Presentation Materials

The use of internal presentation materials specifically *Päällystysteet 2024_yhteenvedo 7.1.2025* and *Vuoden 2024 toteumat ja vuoden 2025 näkymät* also played a role in shaping the new reporting template. These materials provided concrete statistics, such as the achievement of the 4,000 km paving target in 2024, a focus on high and medium traffic roads, and detailed breakdowns of paving costs and ELY-specific performance.

Longitudinal data on pavement conditions and the challenges of maintaining low-traffic networks (PK3), as highlighted in these presentations, were especially valued. They helped provide context for regional disparities and informed long-term planning priorities. Adapting charts, KP-luokka classifications, and historical trend data from internal sources clarified the use of cases for the template Strategic Insights, Paving Status, and Backlog Outlook sections. Stakeholder feedback indicated that these elements helped foster more informed discussions and improved understanding of Väylävirasto's priorities and the complexities of regional infrastructure maintenance.

3.8 Summary of Research Stages and Report Template Design

Early drafts of the report template were created in Microsoft Word to outline the structure and main content areas. These drafts served as a starting point for discussion and were revised several times, with each version tested and refined based on feedback about usability, narrative flow, and visual clarity. As the process continued, it became clear that PowerPoint would be the most practical platform for the final report template, since it is widely used within Väylävirasto and

supports easy editing, sharing, and integration of the agency's visual identity. The final report template combines main visuals and narrative elements on the slides, with presenter notes used to add data storytelling and context for each section. Including the purpose and the objective of the slide, specific context to include and editing tips. Rather than aiming for full compliance with ISO 55000, the focus was on creating a practical, user-friendly reporting tool that makes technical information accessible for non-technical audiences.

To provide a clear overview of the methodological approach and to facilitate the transition to the presentation of results in the findings section, TABLE 3 summarizes the key stages, activities, and sources of evidence that structured this research. This synthesis highlights the iterative and user-centered nature of the process.

TABLE 3. Summary of the Research Process, Key Activities, and Supporting Sources

Stage	What Researcher Did (from thesis)	Primary Sources in Thesis	Purpose / Outcome
1. Problem definition & stakeholder mapping	Framed the practical communication problem; identified key audiences and their recurring questions (internal staff, policymakers, journalists, citizens).	Introduction; Findings 4.1 Stakeholder Needs and Barriers (4.1.1–4.1.4)	Ground the project with real user needs and questions to steer design.
2. Literature review & benchmarking	Reviewed data storytelling, narrative formats, accessibility, ISO 55001, legal frameworks; benchmarked UK ALARM and Ireland TII/SAMP.	Review of Literature (2.1–2.11); International Practices (2.10); International Benchmarking (3.1.1)	Build theoretical foundations and identify transferable best practices.
3. Case study design (Design-Based Research)	Adopted DBR; co-designed with Väylävirasto via iterative prototyping and feedback cycles.	Methodology (3); Case Study Design (3.1); DBR approach (2.6.1)	Integrate theory and practice through iterative, collaborative design.
4. Data collection	Ran semi-structured interviews (9, internal), analyzed internal documents and past presentations; performed international benchmarking.	Data Collection (3.3.4); Selection of Interviewees (3.3.3); Incorporation of Internal Presentation Materials; International Benchmarks in Infrastructure Communication	Gather qualitative and contextual evidence on needs, barriers, and practices.
5. Thematic analysis	Manual thematic coding (Excel); extracted themes: visual clarity, stakeholder needs, communication barriers, tools/capacity.	Data Analysis: Thematic Coding (3.6); Table 2 (Themes & Influence)	Translate raw interview/benchmark findings into design requirements.
6. Prototype development & iterative feedback	Produced Word → PowerPoint template; ran three design iterations; applied feedback on headlines, layouts, visual clarity, accessibility.	Design Case: Väylävirasto Collaboration; Applying Hevner's Three-Cycle Model (3.2)	Evolve the template to be clearer, modular, and reusable.
7. Evaluation & usability checks	Checked completeness, usability, objectivity; aligned with ISO 55001 communication and WCAG accessibility principles.	Evaluating the Template (3.2.1); ISO 55001 (2.9); Accessibility (2.5.2; Legal Frameworks 2.9.1)	Ensure robustness, accessibility, and trustworthiness of the template.
8. Synthesis & recommendations	Summarized findings; delivered guidelines (visual standards, glossary, story brief, data story packs, design lead, Copilot use, expanded benchmarking).	Suggestions (5.1–5.7.5); Conclusion (6)	Provide actionable steps for sustained user-oriented reporting.

4 FINDINGS

In this section, the main findings show that a user-oriented, story-driven approach, using clear visuals and plain language, makes infrastructure data easier for all stakeholders to understand and use. The new template addresses common barriers, but further feedback from external users would be important for future improvements.

4.1 Stakeholder Needs and Barriers

The following findings are based on interviews with Vaylavorasto experts, analysis of common stakeholder questions, and review of previous agency presentations. The researcher identified key stakeholder groups who interact with road and bridge condition data at Vaylavorasto through qualitative analysis of semi-structured interviews and review of relevant literature, including Hamalainen (2025). The primary stakeholders include internal staff (including specialists and non-specialists such as directors, board members, communications, and customer service staff), policymakers, journalists, and citizens, each with distinct roles and information needs.

4.1.1 Internal Staff

Internal staff at Vaylavorasto, including asset management and road condition management experts, as well as board members, directors, communications staff, and customer service staff, play various roles in managing and communicating infrastructure data. While specialists are responsible for handling large datasets and ensuring technical accuracy, other staff, such as directors and communications personnel, focus on sharing information, responding to stakeholder questions, and supporting decision-making. Many internal staff do not have technical backgrounds, and thus rely on clear, high-level summaries and practical visuals rather than detailed metrics. Across all these groups, clarity and consistency of data presentation, as well as the ability to respond quickly to stakeholder needs, are key concerns. One tool that supports both internal and external

information sharing is the Suomen Väylät map and data service, which provides access to a wide range of infrastructure datasets and thematic maps. However, interviews consistently highlight the agency's ongoing struggle to communicate technical information clearly and relevantly to non-expert audiences. While the organization maintains detailed, frequently updated registries of road and bridge conditions, much of this information remains in internal systems or is shared only through ad hoc PowerPoint summaries and Excel files. As one interviewee noted in April 2025, "We report this way today and another way tomorrow. In many cases we must start from scratch."

4.1.2 Policymakers

Policymakers often request summaries and visualizations that show trends, funding allocations, and maintenance backlogs (Väylävirasto specialist, personal communication, July 16, 2025). They usually want clear, big-picture information that helps them make decisions and set priorities. Their questions often relate to budget cycles or political terms, for example: "What is the current value of the maintenance backlog?". They focus on numbers that matter to their region or time in office, such as annual kilometers of road paving or the size of the maintenance backlog (Väylävirasto specialist, personal communication, July 16, 2025). Policymakers rarely request technical details unless these are directly linked to cost or performance outcomes. Visual summaries and strategic overviews are preferred because they are easy to communicate to constituents (Väylävirasto specialist, personal communication, July 16, 2025).

4.1.3 Journalists

Journalists are often interested in trends and regional comparisons. They typically reach out after significant events, such as road collapse or severe weather, and request visuals or maps. Their questions focus on societal impacts, for example: "How does this affect safety in different regions?" Journalists look for quick facts and ready-to-use visuals to include in articles, such as charts and maps that highlight key trends (Väylävirasto specialist, personal communication, July 16,

2025). They prefer concise explanations that make complex data easy to share with the public (Väylävirasto specialist, personal communication, July 16, 2025).

4.1.4 Citizens

Citizens usually ask practical questions about local road quality and safety, such as: “When will my road be fixed?” or “Why hasn’t the snow been cleared in my area?” The specialist from Väylävirasto noted that their concerns are often emotionally charged, especially during winter, when feelings of being neglected are common (Väylävirasto specialist, personal communication, July 16, 2025). As one communications manager explained, “Sometimes the people's opinion, they are pretty emotional because it is their road, it is their daily life. In the wintertime, it is not even about accidents; it is just that they feel they have been abandoned while living in the countryside.” Citizens benefit most from simple explanations and clear maps that make technical information accessible (Väylävirasto specialist, personal communication, July 16, 2025). The following table summarizes the main stakeholder groups, their needs, and how the new reporting template addresses these requirements:

TABLE 4. Summary of Stakeholder Groups, Their Needs, and Template Solutions

Stakeholder Group	Main Needs & Questions	How the Template Addresses These Needs
Internal Staff (specialists, directors, communications, customer service)	- Clear, high-level summaries - Consistent visuals and terminology - Ability to respond quickly to stakeholder questions - Practical tools for sharing information	- Modular, reusable PowerPoint slides - Color-coded charts and simple layouts - Glossary of key terms - FAQ section for common questions - Editable format for easy updates
Policymakers	- Big-picture trends and summaries - Funding allocations and maintenance backlog - Regional comparisons - Actionable insights for decision-making	- Thematic sections (e.g., funding, regional overview) - Strategic overviews and headline indicators - Visual summaries (maps, charts) - Plain-language explanations - Grouped questions by policy relevance
Journalists	- Quick facts and ready-to-use visuals - Regional and temporal comparisons - Societal impacts (e.g., safety, accessibility) - Concise explanations for public communication	- Visual scorecards and annotated charts - Maps highlighting key trends - Short, narrative headlines - Easily shareable visuals - Plain-language summaries for media use
Citizens	- Local road quality and safety - Practical information (e.g., repair schedules) - Simple explanations and clear maps - Emotional concerns about fairness and accessibility	- Region-specific visuals and maps - Plain-language summaries - “Why does this matter?” notes - Glossary for technical terms - FAQ addressing everyday concerns

4.2 Barriers to Effective Communication

Interviews and document analysis identified several factors that can limit the effectiveness of reporting and communication at Väylävirasto. Communication between departments was described as limited, and the absence of a designated report designer sometimes led to inconsistent formats and overlapping work.

A key challenge is the lack of standardized visual templates and reporting guidelines. Interviewees noted that reporting styles, chart formats, and terminology may differ between teams and can also change over time (Väylävirasto specialist, personal communication, April 4, 2025). Without a dedicated report designer, technical experts who often lack formal training in visual communication are responsible for creating charts and graphs (Väylävirasto specialist, personal communication, July 1, 2025). This can lead to variations in presentation and, at times, different information being provided to stakeholders, which may affect clarity and consistency (Väylävirasto specialist, personal communication, April 4, 2025).

4.2.1 Reliance on Individual Expertise

Additionally, the agency's communication processes are heavily dependent on the expertise of individual technical staff. Because internal data systems are not designed for public use, communications staff must rely on these experts to extract and interpret figures, often on short notice (Väylävirasto specialist, personal communication, July 17, 2025). This reliance can create delays and increase inconsistency, as answers may differ depending on who is available to respond (Väylävirasto specialist, personal communication, July 30, 2025). In addition, technical terms such as 'rut depth' or 'International Roughness Index' are often unclear to non-specialists, making it difficult for the public or policymakers to interpret whether a value is positive or negative (Hamalainen et al., 2025). To address this, a glossary of key terms was included at the end of the report template, ensuring that technical concepts were clearly explained.

4.2.2 Challenges with Data Timeliness and Consistency

Another issue relates to the timing of data updates: condition measurements and model outputs are updated on different schedules, so the official value for an indicator can change from month to month (Väylävirasto specialist, personal communication, July 30, 2025). This can lead to confusion and repeated questions from stakeholders. These challenges are not unique to this context. Previous research has observed that inconsistent or unclear information is common in public data portals, and that there is a lack of consolidated practices for publishing data in ways that are accessible, easily understood formats (Oliveira et al., 2023; Kempeneer et al., 2025). Providing data or simple charts might not be enough for the public to understand. Instead, practical communication benefits from narrative context, explanation, and standardized practices to ensure accessibility and comprehension for all audiences.

4.2.3 Limited Use of Storytelling

Although storytelling is widely recognized as necessary for public communication, formal reports and public materials at Väylävirasto remain primarily technical and descriptive, with little focus on narrative framing (Väylävirasto specialist, personal communication, July 17, 2025). Interviewees said current outputs feel like ‘text piled on text about what we are doing’ rather than engaging stories that link technical facts to societal impacts or strategic goals (Väylävirasto specialist, personal communication, July 30, 2025). The lack of narrative structure is seen as a missed opportunity to make complex information more accessible, especially for audiences such as policymakers and the general public.

4.2.4 Reactive Stakeholder Engagement

Communication with external stakeholders is largely reactive, with most inquiries coming in after accidents, winter storms, or budget negotiations (Väylävirasto specialist, personal communication, July 30, 2025). Because there are few reusables, pre-approved report templates, responses are often crafted from scratch

(Väylävirasto specialist, personal communication, April 4, 2025). Additionally, while customer management tools exist, they are not consistently used, leading to reliance on email threads rather than a searchable library of approved answers and visuals. This shows the need for a governance structure that defines which wording and graphics should be the official versions for a set period (Kempeneer et al., 2025). Interviewees expressed a strong preference for modular, reusable templates that could be easily updated and adapted for different audiences. The communications team highlighted the value of templates for ensuring consistency and supporting content reuse across channels such as social media, newsletters, and press releases.

4.3 Structural Obstacles to Public Data Sharing

While templates and standardized formats can support consistency, individual staff members have still encountered obstacles when publishing new content. For example, a specialist attempted to share a recorded webinar but encountered challenges with voice permissions and adding subtitles for accessibility. While these are standard requirements, the process felt overwhelming for the specialists involved. With limited guidance and support, progress slowed and eventually stopped. This shows how bureaucratic steps combined with a lack of communication and assistance can discourage individuals and prevent information from being shared.

Another interviewee described similar frustrations in making maintenance backlog data more accessible to the public. In a conversation on July 4, 2025, a specialist recounted how their team wanted to publish a simple web page to share backlog information. However, they lacked the know-how and time to build a public-facing site themselves, and it was unclear who within the organization could provide support. Although the communications department eventually assisted by setting up the basic web page, the technical team was left responsible for creating all the charts, writing the content, and handling ongoing updates. As a result, the final product was functional but visually unappealing and lacked design consistency. This experience illustrates how limited support and fragmented processes can hinder the effectiveness of data communication efforts, even when

the motivation to share information is strong (Väylävirasto specialist, personal communication, July 4, 2025).

4.4 Use of Suomen Väylät as a Public Information Channel

One method Väylävirasto uses to share infrastructure information with stakeholders is the Suomen Väylät map and data service. While not a formal report, Suomen Väylät serves as a digital platform for making road, rail, and waterway data accessible to the public. This approach supports transparency by allowing users to explore up-to-date datasets and thematic maps outside of traditional reporting cycles. However, interviews and feedback indicate that, despite its potential, the service has proven difficult for many non-expert users to navigate and interpret. Stakeholders noted that the technical nature of the interface and the lack of clear guidance or narrative context can make it challenging for citizens and even some professionals to extract meaningful insights. As such, while Suomen Väylät represents an important step toward open data and proactive stakeholder engagement, it does not replace the need for structured, audience-specific reports. Instead, it highlights the ongoing challenge of ensuring that digital tools are both accessible and understandable to all intended audiences.

4.4.1 Insights from Interviews and Feedback

Visuals such as maps, simple bar charts, and infographics were highlighted as the most effective formats for both policymakers and the public. As one specialist from Väylävirasto observed, while a variety of chart types can enhance engagement, prioritizing simplicity ensures that the visuals remain accessible and effective for all audiences. This perspective directly informed the design rules for the template, guiding the selection and presentation of visual elements to balance diversity with ease of understanding. As one team member explained, “The communication team's strategy relies heavily on repetition of key messages. These are shared across multiple channels, social media, press releases, internal responses, and customer service to avoid confusion” (Väylävirasto specialist, per-

sonal communication, July 17, 2025). One specialist stated, “Stakeholders respond best to infographics, short videos, and narrative visuals that simplify complex data without technical overload” (Väylävirasto specialist, personal communication, April 4, 2025). Typography also plays a quiet but powerful role. Interviewees said the value of visuals that are “really clean,” and noted that “simple is best,” favoring layouts that focus on one chart per insight.

4.4.2 Design Choices and Accessibility

Design elements such as color choices, font hierarchy, and the use of annotations play a significant role in how non-expert audiences interpret infrastructure data (Shao et al., 2024). For example, slide 5 in the report template uses color-coded condition classes, blue, yellow, red, and dark red, to visually communicate disparities. Stakeholders noted that these “traffic light” colors make it easy for viewers to quickly see what is important and interpret where urgent action is needed, while keeping the presentation clear and easy to understand. As feedback continued, color-coded scorecards were added, so users could quickly see the condition of roads and bridges without having to interpret detailed charts. Features such as bold callouts and concise text annotations further clarify key points by providing brief narratives. At the same time, short summaries effectively highlight urgent issues without requiring the audience to interpret complex raw data (Shao et al., 2024). These design decisions are guided by the Web Content Accessibility Guidelines, which aim to reduce cognitive load and improve usability (W3C WCAG Guidelines, 2023).

In addition, regulatory frameworks such as the EU Accessibility Act (Directive (EU) 2019/882) establish legal requirements to ensure that digital products and services, including those provided by government agencies, are accessible to all audiences, especially users with disabilities (European Union, 2019).

4.4.3 Iterative Design

A continuous, collaborative, and iterative process guided the development of the stakeholder-oriented report template for Vöylävirasto. Early feedback highlighted the need to simplify technical jargon to make key messages more understandable. As the design evolved, stakeholders requested more visuals, such as regional maps and trend charts, to help users quickly understand patterns in road and bridge conditions. One of the most common suggestions from stakeholders was to focus each visual or slide on a single key message, using everyday language and clear headlines to guide the audience's attention. Another effective technique is narrative framing, pairing visuals with explanatory headlines or calls to action that link the data to real-world concerns. As one advisor noted, "It is recommended to start with the main messages, so stakeholders know what to focus on" (Vöylävirasto specialist, personal communication, October 13, 2025). In response, each slide in the report template begins with a clear, action-oriented headline that summarizes the main message, ensuring that key insights are immediately visible (e.g., slide 6: 'Funding Shortfall Reaches 1.2 Billion...'). Alongside these headlines, the report template features plain-language summaries and simplified explanations of technical terms. A "why does this matter" section in the presenter's notes to further highlight the data's real-world significance, ensuring the key messages resonate with a broader audience.

4.4.4 Report Template Format Selection

The initial report template drafts were created in Microsoft Word to outline the structure and identify the report's content areas. These early drafts served as a roadmap, which was revised through multiple versions as the order of sections and the overall flow of information were experimented with. Together with Vöylävirasto's staff, it was decided that the final version of the template would be developed in PowerPoint. This decision was based on the platform's practicality and accessibility, as it is the most widely used tool within Vöylävirasto. PowerPoint enables easy editing, sharing, and presentation by staff members, and supports the integration of the organization's official images and logo to ensure consistency with its visual identity. PowerPoint was particularly effective in combining

the main visuals and narrative elements directly on the slides. At the same time, the presenter's notes were used to include data storytelling components and a narrative of the information on the slide. The presenter notes on these slides reinforce the main takeaway by explaining the implications for travelers, planners, and policymakers, making the slides clear and easy to read, and providing guidance on how to tell a story with the data so it is more meaningful for stakeholders. This approach made the slides clear and easy to read, while also guiding how to tell a story with the data to make it more meaningful for stakeholders. A key takeaway from this process was that staff were mainly interested in how the report told a story, not just the data or text explaining the graphs. This insight shaped many design choices. To make the information more relevant, specific impacts and affected user groups were explained, and comparisons over time and across regions were added to show changes and trends.

4.4.5 Effectiveness of the New Report Template

Stakeholders suggested grouping questions, resulting in the creation of a Frequently Asked Questions (FAQ) section at the beginning of the report template that summarizes the primary questions the report aimed to answer. Internal stakeholders noted that presenting all the key questions on a single slide, such as “Which regions have the worst roads and bridges?” and “How has the condition of roads and bridges changed over time?”, felt overwhelming and made it difficult to quickly grasp the report’s structure. (see FIGURE 10, Road and Bridge Network Condition Report, 2025, slide 5).

Which regions have the worst roads and bridges?	How has the condition of roads and bridges changed over time?	Why do some regions suffer more deterioration than others?
What is the size and trend of the maintenance backlog for roads and bridges?	How does the funding gap (budget vs. need) affect maintenance and repairs?	How has the funding situation changed over the past decade?
What are the policy implications of backlog growth and funding shortfalls?	How much is being spent on road and bridge maintenance?	How does preventive maintenance compare to emergency repairs in terms of cost and effectiveness?
Which regions are prioritized for investment and why?	How much of the network is managed by Vaylätirasto vs. private/forest roads?	How are road and bridge conditions measured (rutting, evenness, visual inspection)?
How often are roads and bridges inspected?	What are the key performance metrics (rutting depth, surface evenness, IRI)?	How can I report a problem or give feedback about roads and bridges?

In this presentation you will find the answers to:

5

FIGURE 10. FAQ-style introduction slide listing all key questions addressed in the report template. Adapted from *Road and bridge network condition report* (unpublished PowerPoint presentation). Created by the author.

In response to this, these questions were divided into thematic sections, such as regional overview, condition trends, maintenance and funding, policy and investment, and public engagement. This change was seen as making the information more visually organized and accessible, helping users to navigate the content more easily and focus on the areas most relevant to their interests. (see FIGURE 11, Suoninen, 2025, slide 4).

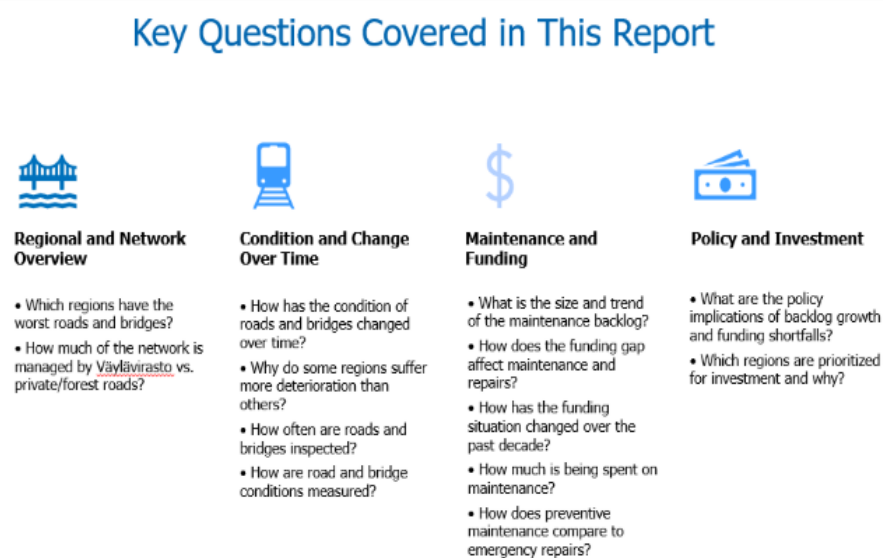


FIGURE 11. Grouped FAQ slide showing thematic sections for key report questions. Adapted from *Road and Bridge Network Condition Report* (slide 4), Unpublished PowerPoint Presentation). Created by the author.

4.5 Generalizability and Limitations

This study examines the communication challenges faced by VÄylÄvirasto as a case through which to explore broader questions about public data storytelling and stakeholder-focused communication. While the study provides in-depth insights into the development of stakeholder-oriented reporting within VÄylÄvirasto, it is important to note that the findings may not be fully generalizable to other organizations or contexts, given the focus on a single agency, specific location,

and the reliance on internal stakeholders for feedback. There are two main reasons for this. First, Väylävirasto did not require or request interviews with external users. The project was planned together with the agency, and their priority was to focus on staff and internal experts at this stage.

Second, this report is a template, not a final product. It is a template showing how future reports could look. The report template uses mock data and example visuals to illustrate design and structure. The report is intended to guide and inspire future work rather than provide definitive conclusions. Although it includes preliminary recommendations, these are exploratory. Because the template has not yet been applied in operational contexts, external feedback was considered less critical at this stage. The main goal was to ensure the report template meets Väylävirasto's needs and standards. To address this limitation, the study benchmarked Väylävirasto against international best practices to broaden the relevance of its findings.

In addition, relying only on internal interviews introduces the potential for bias, since perspectives from external stakeholders were not included in the primary data collection. This means the findings may reflect a limited view shaped by the organization's own priorities and assumptions. Internal staff often share similar experiences and values, which can lead to a narrow understanding of the issue and may overlook challenges or needs that are more visible to external audiences (Tarnoki & Puentes, 2019). Including external voices could provide a more balanced picture, reduce the risk of confirmation bias, and help identify blind spots that internal stakeholders might not recognize (Tarnoki & Puentes, 2019). For this study, the decision to focus on Väylävirasto staff was based on project scope and agency priorities, but it is important to acknowledge this limitation and suggest that future research should consider broader stakeholder inclusion to enhance generalizability and reduce potential bias.

4.6 Regional Disparities in Infrastructure Conditions

Regional disparities in road and bridge conditions are important information for all stakeholders, as they directly affect travel, safety, and resource allocation across Finland. The slides "Northern and Rural Regions Have the Poorest Roads"

(slide 5), “Regional Priority Mapping” (slide 6), and the regional trend chart (slide 19) provide actionable insight by making these disparities both visible and comparable. Slide 5’s color-coded bar chart quickly identifies regions with the highest share of roads in poor condition, while slide 6 links these findings to maintenance backlog and investment urgency. The trend chart on slide 19 shows how the length of poor roads has changed over time in each region, highlighting that deterioration is accelerating most rapidly in areas such as Etelä-Pohjanmaa, Lapland, and Pirkanmaa. The presenter notes clarify the underlying causes and implications: Uusimaa, Pirkanmaa, and Varsinais-Suomi are under the most significant strain due to high traffic and growing backlogs; Lapland and Kainuu face persistent challenges from long networks and harsh weather; and central regions like Keski-Suomi and Pohjois-Pohjanmaa experience moderate backlog growth, reflecting the influence of funding, asset age, and climate. These narratives personalize national data and invite policy dialogue on whether to prioritize high-traffic efficiency or regional fairness.

4.6.1 Making Technical Metrics Understandable

Kosara and Mackinlay (2013) argue that compelling storytelling in public reporting requires weaving together words, images, and numbers to convey impact. In line with this, the report template translates technical indicators into accessible narratives, helping stakeholders grasp the real-world implications of infrastructure data. For instance, metrics such as rutting, International Roughness Index (IRI), and backlog are made meaningful by linking them to outcomes like safety restrictions, freight delays, and user satisfaction, clarifying why the data matters, not just what it measures. Presenting the maintenance backlog as a cumulative trend line, rather than isolated figures, further illustrates the compounding nature of deferred maintenance (Slides 8–9), while the use of metric panels (Slides 15–16) reinforces this communicative goal by pairing definitions with icons and concise “why it matters” statements. To support transparency and understanding, the report template also explains how condition data is collected, using automated sensors, LiDAR-equipped vehicles, and visual surveys, and details the frequency of inspections (Slides 15–19), ensuring stakeholders can follow both the meaning and the methods behind the reported figures.

4.7 Summary of Key Findings

Interviews, stakeholder feedback, and past reporting practices show that the agency has had trouble making technical infrastructure data easy to understand and useful for stakeholders. This issue highlights the need for standardized formats, which Vöylävirasto staff see as key for accuracy and consistency. To meet these needs, the new report template uses a modular, question-based structure that groups information into clear sections, making it easier for readers to find what matters most. Design features like color-coded visuals, short annotations, and plain-language summaries also help make complex data more approachable for non-experts.

Key findings point to both barriers and opportunities. Stakeholders including internal staff, policymakers, journalists, and citizens struggle with inconsistent reporting formats, heavy technical jargon, and a lack of narrative context that connects data to everyday concerns. Internal processes often depend on individual expertise, which leads to delays and differences in how information is shared. Citizens, in particular, raise emotional concerns about road conditions, showing the need for communication that is not only clear but also empathetic. Policymakers and journalists need concise, actionable summaries and strong visuals to support decision-making and public discussion. Without standardized templates and guidelines, communication has become fragmented, and materials are rarely reused. Digital platforms like Suomen Vöylät offer potential, but they remain difficult for non-experts to navigate.

The new template's modular structure, thematic organization, and FAQ-style introductions directly address these challenges by making information more visually organized and accessible. Color-coded visuals and concise annotations help users quickly identify key issues, while plain-language summaries and glossaries clarify technical terms. The report template also incorporates recognized standards, such as the SFS Finnish Standards and W3C accessibility guidelines, and emphasizes transparency through explicit references and open feedback channels. By openly sharing data sources and methods, the report template reduces the risk of bias and builds trust with readers. Regional disparities in infrastructure

conditions are made visible through targeted visuals and narrative framing, linking technical metrics to safety, resource allocation, and policy priorities. Iterative development, guided by continuous stakeholder feedback, has resulted in a report template that balances clarity, consistency, and adaptability for future digital formats. This approach ensures that Väylävirasto's reporting remains relevant, trustworthy, and responsive to evolving stakeholder needs, while supporting ongoing improvement and engagement.

TABLE 5. Key Findings Table

Challenge / Opportunity	Description	Impact on Stakeholders
Technical & Inconsistent Communication	Road and bridge data is often presented in technical language and inconsistent formats.	Non-experts (policymakers, journalists, citizens) struggle to understand and use the information.
Need for Clear, Visual Summaries	Stakeholders require accessible, visual, and plain-language summaries.	Improves decision-making and public engagement.
Lack of Standardized Templates	No unified report templates or guidelines.	Leads to confusion, repeated work, and inconsistent answers.
Reliance on Individual Expertise	Reporting depends on individual staff knowledge.	Causes delays and variability in responses.
Missing Storytelling & Narrative Framing	Reports lack narrative context and storytelling.	Limits public engagement and understanding.
Complexity of Digital Tools	Tools like Suomen Väylät are available but too complex for non-experts.	Reduces accessibility and usefulness for the general public.
Modular, Question-Based Templates	New templates group information by stakeholder questions.	Improves clarity, consistency, and accessibility for all users.

To address these challenges, effective reporting requires more than just presenting numbers it calls for telling a clear, structured data story. The following figure illustrates a step-by-step approach to transforming raw infrastructure data into an engaging narrative. This process begins with defining the audience and purpose, then moves through organizing content, visualizing key insights, and framing the message in plain language. By following these steps, reports can shift from technical documents to tools that inform decisions, build trust, and foster public understanding.



FIGURE 12. How to Tell a Data Story. Created by the author using Microsoft Copilot.

5 SUGGESTIONS

The following suggestions address the key challenges identified in Väylävirasto's current reporting and knowledge-sharing practices. Drawing on both research evidence and practical observations, these recommendations aim to enhance clarity, consistency, and accessibility in infrastructure communication, ensuring that information is both actionable and inclusive for all stakeholders.

5.1 Visual Guidelines

Visual hierarchy shapes how readers process information. To make organizational reporting more transparent and consistent, standardized visual guidelines are recommended. Currently, accessibility and visual checks are managed by external partners, and staff are generally satisfied with this arrangement. However, as reporting needs grow, Väylävirasto should consider developing supplementary internal guidelines and structured training to reduce reliance on external support and ensure all staff are equipped to meet accessibility standards. These guidelines should include practical details such as font types and sizes, color schemes, logo placement, and spacing rules so that all presentations and documents follow the same structure (Naegle, 2021). Väylävirasto's report templates should apply distinct typography for headings, subheadings, and body text, along with consistent spacing and alignment. This structure ensures key insights, such as callouts and contextual notes, stand out first (Shao et al., 2024). Research on effective slide design highlights the importance of limiting text, using high-contrast colors, and organizing content so that the most important points stand out (Naegle, 2021). Including these principles in a report template gives staff a clear reference, reduces confusion, and saves time when preparing reports. Consistent visual standards also help maintain a strong brand identity and make information easier to understand, especially when different departments produce materials (Brumberger, 2007).

Color coding is effective for status indicators. The researcher recommends using green for positive, amber for caution, and red for critical. Pair colors with text or icons and ensure contrast to meet WCAG 2.2 requirements. In addition to visual

consistency, accessibility should be a priority. Training staff on visual communication and accessible design ensures they know how to apply effective practices, such as choosing readable fonts, checking color contrast, and adding alternative text for images (Drye et al., 2023). These steps follow recognized accessibility guidelines and show a commitment to clear and inclusive communication (American Alliance of Museums, 2021).

5.1.1 Making Terms Clear

Technical reports at Väylävirasto often contain specialized terminology that can be challenging for stakeholders to understand. While the current practice of having external partners review reports for clarity and accuracy is valuable, it does not fully address the need for accessible communication. A more effective approach would be to integrate plain-language summaries and glossaries directly into report templates from the beginning. Glossaries provide standardized definitions for key terms, reducing ambiguity, and supporting consistent communication across documents. Incorporating these elements into various formats, such as reports, presentations, and online views, would make information easier to understand, reduce confusion, and enable stakeholders to make better-informed decisions. Research emphasizes that clarity and consistency are essential for effective communication and that terminology management plays a critical role in achieving these goals (ISO, 2023; Stoll et al., 2022). Additionally, digital tools like Copilot could further streamline this process by generating glossaries from existing content, suggesting simple explanations for complex terms, checking for jargon, and ensuring consistency across all documents.

5.2 One-Page Story Brief Before Slide Creation

A one-page Story Brief is a simple planning practice to complete before creating any slides for a presentation. Its main purpose is to help everyone involved agree on what the presentation is about and how the story will be told. By starting with a structured brief, presenters can avoid the common mistake of making slides that don't fit together or tell a clear story (Segel & Heer, 2010). Research shows

that planning the story first makes it easier for the audience to understand and stay interested, especially when the topic is technical or involves a lot of data (Shao et al., 2024). Väylävirasto could include practical examples and templates in its reporting process, and Copilot could help by making it easier to draft and review these briefs. This would help staff create clear, focused presentations that are easy for the audience to follow.

5.3 Core Elements of the Story Brief

The brief should be concise, ideally one page, and include the following five components:

1. Audience

Identify who the presentation is for and what matters to them. Consider their level of technical knowledge and expectations.

Example: Policymakers may need a high-level overview of project benefits, while bridge engineers require detailed structural data.

2. Goal

Define the desired outcome or decision that the presentation should influence.

Example: Secure approval for a new bridge maintenance program or obtain funding for a highway expansion project.

3. Key Message

State the single most important idea you want the audience to remember.

Example: “Upgrading the Tampere–Helsinki corridor will reduce travel time by 15% and improve safety for 50,000 daily commuters.”

4. Supporting Data Sources

List the evidence that reinforces your key message.

Example: *Traffic flow simulations, accident statistics, lifecycle cost analysis, and environmental impact assessments.*

5. Figure Inventory

Outline the visuals you plan to include, charts, maps, diagrams, and link them to the story flow.

Example: *A map of proposed road widening sections, a bar chart comparing maintenance costs, and a timeline for bridge reinforcement phases.*

Visuals can enhance a presentation but pairing them with a clear narrative further strengthens the message. Embedding data within a story helps the audience understand and engage with the information effectively (Segel & Heer, 2010). Starting with Story Brief ensures that each slide contributes to a cohesive and compelling overall narrative.

5.4 Data Story Packs

The researcher recommends that Väylävirasto develop and offer a set of ready-mades, downloadable data story packs within the Suomen Väylät portal and/or on the website. Each pack should focus on a specific theme, such as a “Regional Condition Snapshot,” a “Backlog Explainer,” or “Bridges at a Glance.” For example, the Regional Condition Snapshot could include a visual map, three key bullet points, and a short FAQ to summarize the current state and disparities in road and bridge conditions across regions. “The Backlog Explainer” could present a trendline chart, a plain-language definition, and a summary of societal impacts to help stakeholders understand what the maintenance backlog is, why it matters, and how it is changing. “Bridges at a Glance” could provide key counts, condition bands, restrictions, and freight impacts, offering a concise overview of bridge infrastructure and its implications. Each pack should feature plain-language annotations and a brief story abstract to make the information accessible and meaningful for non-expert audiences (Dahlstrom, 2014; Grafl & Schuler, 2022).

These packs would be modular and pre-approved, designed for use and sharing across channels: intranet updates, dashboard banners, PowerPoint templates,

portal highlights, social media snippets, and press lines. Adding these themes to the packs and sharing them on all channels provides stakeholders with reliable, up-to-date information.

5.5 Designate a Reporting Design Lead

While Väylävirasto ensures compliance with technical and accessibility standards for public reporting, a recent interview clarified that there is no dedicated internal role explicitly focused on the content and narrative structure of reports (Väylävirasto Specialist, November 2025). Visual and technical support is available through the communications team, which assists with design and accessibility aspects. External partners currently manage reporting standards. However, questions about crafting a clear, audience-appropriate story or choosing the most effective data visualizations may not have a clear point of contact. Appointing a Reporting Design Lead would address this gap. This role would provide staff with guidance on both content and design, ensuring best practices are consistently applied throughout the organization.

This challenge extends to visualization choices, which play an important role in how information is understood. Research shows that chart selection should reflect the user's purpose and level of expertise, rather than relying on generic templates (Shao et al., 2024). A design lead can help teams choose chart types that improve clarity and reduce confusion. Governance frameworks for reporting platforms also stress the need for defined roles and approval processes to maintain trust and consistency (SFS Finnish Standards, 2025). This aligns with creating a role that manages templates and oversees a simple approval workflow for public reports.

5.5.1 Rotational Approach

To build shared expertise, Väylävirasto could implement a 6-month rotation for the Reporting Design Lead role. Given that knowledge sharing at Väylävirasto is

largely informal and dependent on regular meetings and personal contacts, implementing a rotational approach for the Reporting Design Lead role would help distribute expertise more evenly and encourage collaborative learning across teams. This would allow team members to take turns leading on visual hierarchy, narrative headlines, color/contrast, and glossary consistency. Each rotation could include maintaining the approved template library and ensuring accessibility standards. Rotating the role would share responsibility and follow effective practices in workplace visual communication, as Brumberger (2007) notes that design decisions benefit from diverse perspectives and collaboration. Such an approach promotes cross-team learning, keeps design practices up to date, and fosters a culture in which visual clarity and accessibility are treated as shared organizational priorities.

Key Responsibilities

- **Provide Guidance:** Act as the go-to person for questions on color hierarchy, chart selection, and accessibility.
- **Manage Templates:** Maintain and update a library of approved report templates.
- **Follow Reporting Trends:** Monitor new reporting and accessibility trends and share updates with the team.
- **Review Reports:** Oversee a basic approval process to ensure clarity and compliance.
- **Support Learning:** Offer short training sessions and practical tips for staff.
- **Using Copilot in User-Oriented Infrastructure Reporting**

To advance Väylävirasto's goal of producing user-oriented infrastructure reports, Microsoft 365 Copilot can be a very helpful tool. Copilot makes it easier to automate both the writing and the visual parts of a report. For example, it can search for information from within the organization or from public sources, and it can provide answers in different formats, including links to the data sources (Microsoft, 2025a). If the information needed is not online, staff can simply upload files like PDFs, Excel sheets, or Word documents, and Copilot can then summarize the content, answer questions, or create charts and graphs based on the data (Microsoft, 2025b; Dhiman, 2024). Copilot also works with Power BI, which means it can help turn complex maintenance and asset data into clear, useful

insights for different audiences (Microsoft, 2025a). This could be especially valuable for topics like maintenance backlogs, asset condition trends, and policy impacts.

Interviews show that Copilot and similar AI tools are used only occasionally at Vöylävirasto and are not yet a regular part of the workflow (Vöylävirasto Specialist, November 2025). Most of the time, Copilot is used for simple tasks such as structuring or summarizing reports, and only a few departments are experimenting with AI projects. To get the most out of Copilot, the researcher recommends that Vöylävirasto start with small pilot projects, offer training, and create clear use cases for staff. This way, people can move from just experimenting with Copilot to using it as a normal part of their work, making reporting more efficient and consistent across the organization.

5.6 Visual Creation with Copilot

Copilot's natural language interface enables users to generate visuals by describing their needs in everyday language (Microsoft, 2025a). For example, when reporting on the road maintenance backlog, an author can request an up-to-date graph showing how the backlog has changed over the past decade or an annotated map highlighting regions with the highest proportion of roads in poor condition (Microsoft, 2025a). Similarly, spider charts can be created to compare performance against established targets, supporting benchmarking and strategic planning (Ahonen et al., 2016). These features, however, depend on the availability of accurate and valid data from online sources or from user-uploaded files, such as spreadsheets or reports (Microsoft, 2025b; Dhiman, 2024).

The Copilot in Power BI allows users to create and refine visuals through natural language prompts. It works by interpreting the underlying semantic model, which defines relationships and measures within the dataset, to select relevant fields and configure charts. For example, a prompt such as "Create a bar chart comparing annual maintenance budgets to estimated needs" results in Copilot generating a suitable visual ready for inclusion in a report template or dashboard (Microsoft, 2025a). Users can then issue follow-up prompts like "Add a slicer for

region” or “Change this to a stacked bar chart.”. To ensure accurate results, Vöylävirasto should prepare AI-ready semantic models by applying clear naming conventions, defining relationships, and adding synonyms for key infrastructure and maintenance terms. For example, fields such as annual maintenance budget, estimated repair needs, and road segment condition should use standardized terminology and include alternative names commonly used in internal reports. These practices will help Copilot interpret prompts.

5.7 Expanding International Benchmarking

Benchmarking leading international reports helps Vöylävirasto keep its infrastructure communication current and user-friendly. Examples such as the UK’s ALARM report and Ireland’s Strategic Asset Management Plan demonstrate effective practices through plain language, clear visuals, and modular structures that make information easy to navigate. Adopting this approach means looking at how these reports organize their content, use color and layout, and explain technical terms. Vöylävirasto can then adapt the best ideas to fit its own needs and the expectations of Finnish stakeholders. For example, grouping questions by theme, using color-coded charts, and including short summaries can make reports easier to read and understand.

To get the most benefit from benchmarking, Vöylävirasto should make sure that insights from these international examples are shared systematically across teams, not just through informal conversations. Once these insights are captured, the next step is to make them easily accessible and actionable for staff. Copilot can help by searching for and summarizing key findings from benchmarking reports, making it easier for staff to learn from global effective practices. For instance, staff could use a prompt like, “Find recent international infrastructure reports that use plain language and strong visual design, and summarize their main features.” This would support continuous improvement and help all users stay up to date with new ideas. Beyond the UK’s ALARM report and Ireland’s Strategic Asset Management Plan, several other international reports offer valuable insights that Vöylävirasto can draw from to strengthen its approach.

5.7.1 2023 Condition Monitoring Report (ThoughtLab/Bentley Systems)

The 2023 Condition Monitoring Report, authored by ThoughtLab and Bentley Systems, provides an overview of how organizations worldwide are adopting digital technology to monitor infrastructure assets. The report explains how automated systems and real-time data can replace manual inspections, making processes faster and more efficient (ThoughtLab & Bentley Systems, 2023). It shows how combining sensor data and asset records creates a complete, transparent view of infrastructure health. For Väylävirasto, this example demonstrates how digital monitoring and clear visualizations can replace manual processes and improve decision-making (ThoughtLab & Bentley Systems, 2023).

5.7.2 Infrastructure Monitor (2024). – Executive Summary (World Bank/PPIAF)

The Infrastructure Monitor 2024, published by the World Bank and Public-Private Infrastructure Advisory Facility (PPIAF), is notable for its clear organization and narrative that guide the reader through key trends in infrastructure investment (World Bank & PPIAF, 2024). Instead of simply listing data, the report breaks down complex topics such as investment gaps and regulatory changes into easy-to-follow sections, using straightforward language and visual aids to explain why these trends matter.

5.7.3 NSW Infrastructure Digitalisation and Data Policy (Infrastructure NSW, 2025)

The NSW Infrastructure Digitalisation and Data Policy, produced by Infrastructure NSW, is a strong example of how to set clear standards for managing infrastructure data across its lifecycle (Infrastructure NSW, 2025). The policy provides step-by-step guidance on creating a standard data model, setting information requirements, and ensuring data is easy to share and reuse. It also emphasizes the importance of accessibility, staff training, and compliance with international

standards such as ISO 19650 and ISO 55001. For Väylävirasto, this report is a useful benchmark for developing internal guidelines, improving reporting consistency, and ensuring data can be used effectively across teams and projects (Infrastructure NSW, 2025).

5.7.4 ROMA: State Assessment of Road Markings in Denmark, Norway, and Sweden (NordFoU, 2019)

The Road Markings Assessment (ROMA) report, coordinated by NordFoU, is a good example of cross-country benchmarking practice. It uses a consistent set of indicators to compare road marking quality in Denmark, Norway, and Sweden, making it easy to see differences and similarities (Vadeby et al., 2020). The report's transparent methodology and use of visual summaries help make technical results understandable for a broad audience. Väylävirasto could benefit from adopting similar benchmarking methods, using standardized measurements and regular assessments to track progress and support transparency. The ROMA report also shows how straightforward explanations and visual tools can make even technical findings accessible to both specialists and the general public (Vadeby et al., 2020).

5.7.5 Summary of Suggestions

In summary, these suggestions offer Väylävirasto a practical roadmap for enhancing the clarity, consistency, and accessibility of infrastructure reporting. By adopting modular templates, establishing systematic processes, and embedding plain-language glossaries, the organization can streamline knowledge sharing and make technical information more approachable for all stakeholders. The introduction of data story packs and standardized visual guidelines will further support transparent communication, while designating a Reporting Design Lead, ideally through a rotational approach, ensures that good practices are consistently applied and expertise is shared across teams.

Utilizing tools like Microsoft Copilot and benchmarking against international standards will help Väylävirasto maintain effective reporting, making it easier to produce user-oriented materials that inform both internal and external audiences. Implementing a one-page Story Brief before slide creation will keep presentations focused and engaging, and ongoing review of global best practices will foster continuous improvement.

Together, these recommendations lay the foundation for a more proactive, inclusive, and reliable reporting culture, one where information is not only accurate and up to date but also easy to understand and act upon. Designing report content around FAQs and thematic mapping of stakeholder needs will further ensure that each section of the report template answers real user questions and aligns with the most pressing topics, supporting more targeted and effective communication. By implementing these recommendations, Väylävirasto will strengthen its ability to communicate clearly, enabling stakeholders to make informed decisions and foster meaningful engagement with the public.

6 CONCLUSION

This thesis aims to address a persistent challenge in public sector communication: how to make infrastructure reports technically accurate and genuinely useful and accessible for the diverse stakeholders who rely on them. Through a case study with Vöylävirasto, the research explored four key questions:

1. Can a user-oriented reporting template improve the accessibility and usefulness of infrastructure data for different stakeholder groups?
2. Does structuring reports around frequently asked stakeholder questions enhance clarity and relevance?
3. Do data storytelling techniques, such as clear visuals, narrative headlines, and plain-language summaries help transform technical data into actionable stories?
4. Can international best practices be adapted to improve the clarity, consistency, and accessibility of Finnish public sector reports?

The findings demonstrate that the way data is presented is as important as the data itself. By co-designing a modular, question-based reporting template with Vöylävirasto staff and benchmarking against leading international examples, this thesis delivers both a practical report template and a set of guidelines for future adaptation. The report template's use of color-coded visuals, grouped questions, and concise summaries proved effective in making complex topics more accessible to internal staff, policymakers, journalists, and citizens alike.

However, the research has limitations. The focus on internal stakeholders means that some external perspectives may not be fully captured, and the template itself has not yet been tested in official reporting cycles. Future research should broaden stakeholder inclusion, pilot the template in diverse contexts, and evaluate its impact through structured feedback from citizens, journalists, and policymakers. Additionally, as digital tools and user expectations evolve, integrating AI-powered solutions such as chatbots offers potential to further enhance accessibility and stakeholder engagement.

Improved infrastructure reporting has broader societal benefits. By making technical information transparent and relatable, public agencies can foster greater trust, support evidence-based policymaking, and empower citizens to participate in infrastructure debates. The principles and practices outlined in this thesis provide a foundation for continuous improvement in public sector communication, hopefully setting a new standard where infrastructure data is accurate and actionable for all users.

Data storytelling is important because it bridges the gap between raw technical information and meaningful understanding for diverse audiences. For Väylävirasto staff, data storytelling is a communication technique and a mindset shift that transforms how reports are created and how their impact is measured. Instead of beginning with a collection of charts, tables, and technical details and then trying to assemble them into a presentation, effective reporting starts by defining the story the presenter wants to tell: What is the key message for this audience? What decision or action should be supported? Once the narrative is clear, such as highlighting the urgency of maintenance backlogs in a specific region then the supporting data, visuals, and sources can be selected to reinforce that story. This approach ensures that every slide, chart, and summary is purposeful and directly answers stakeholder questions. All in all, by focusing first on the narrative, Väylävirasto staff can make reports more actionable and help policymakers see the specific impact of their decisions, enabling citizens to understand how infrastructure affects their daily lives.

Key Takeaways:

- User-oriented design and data storytelling make infrastructure reports more accessible and relevant for all stakeholders.
- Modular templates and clear visuals support consistency, transparency, and trust in public sector communication.
- Stakeholder feedback and international benchmarking are essential for developing practical, adaptable reporting tools.
- Future work should expand stakeholder inclusion, pilot new digital tools, and evaluate real-world impact.

Better reporting empowers decision-makers and citizens, strengthening engagement with infrastructure issues.

6.1 Future Research

Future research should focus on using the developed reporting template in operational cycles, allowing for the measurement of comprehension, recall, and reuse among different stakeholder groups. It is important to include a broader range of stakeholders, such as citizens, journalists, and policymakers, in subsequent evaluations to compare needs across regions and varying levels of data literacy. Comparative studies in other agencies or countries would help to test the transferability and adaptability of the template.

Additionally, there is a lot of potential in using artificial intelligence tools, like Copilot agents, to help with reporting. These tools could make things easier by automatically sorting FAQs, creating glossaries, and even making visual summaries. Evaluating the accessibility gains from such digital enhancements would be a valuable direction for future inquiry. Research should also examine governance and workflow aspects, including approval processes, template stewardship, and the design of roles such as a rotational Reporting Design Lead.

Formal accessibility audits, including WCAG 2.2 compliance checks and screen-reader testing, are recommended to ensure that published reports meet the highest standards of inclusivity. Finally, future studies should examine how improved reporting practices influence policy and funding decisions over time, in order to provide evidence of the broader societal benefits of user-oriented infrastructure communication.

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Appendix 1. Report Template

Slide 1.

Road and Bridge Network Condition Report

Understanding Regional Gaps
and Funding Challenges

17.12.2025



Presenter Notes:

Purpose: This slide sets the stage for the presentation, clearly stating the topic and the main challenge: understanding regional gaps and funding issues in Finland's road and bridge network.

How to present: Emphasise that the presentation is designed to answer real stakeholder questions, not just present technical data.

Edit tips: Keep the subtitle concise and relevant to the audience. If the audience is Pirkanmaa-focused, add a line in the subtitle: "with focus on Pirkanmaa".

Ensure date is current.

Use a clean, and bold font

If presenting to a specific region or group, tailor the subtitle accordingly.

Slide 2.

Agenda



	Regional Differences in Road and Bridge Conditions
	Funding Gaps and Maintenance Backlog
	Investment Priorities and Trends
	Network Overview (Roads & Bridges)
	Assessment Methods & Key Metrics
	Policy Implications & Recommendations
	Feedback & Contact Information

2

Presenter Notes:

Purpose: The agenda uses a modular, thematic structure to help audiences navigate the presentation. This approach, supported by your thesis, reduces cognitive load and sets clear expectations.

How to present: Walk through each agenda item, briefly explaining what will be covered.

Link each section to a stakeholder need (e.g., “Regional differences for planners, funding for policymakers, assessment methods for technical staff”).

Edit tips: Limit to 6–7 agenda items for clarity.

Use icons or color coding to visually group related topics.

Ensure the agenda matches the actual slide sequence.

Slide 3.

Regional Disparities in Road and Bridge Conditions Are Increasing



Infrastructure Condition Changes

Significant changes in road and bridge conditions highlight regional disparities in maintenance and deterioration rates.

Backlog and Funding Shifts

Shifts in backlog and funding allocations affect future maintenance and infrastructure improvement efforts.

Project Coverage and Activity

Recent projects focus on pavement and bridge programs, showing active efforts to address infrastructure needs.

Performance Metrics and Targets

Annual targets and thrice-yearly condition updates ensure strategic prioritization and timely assessment.

3

Presenter Notes:

Purpose: This slide frames the core problem: significant changes in infrastructure condition, shifting backlogs, and funding allocations.

The thesis recommends starting with a clear problem statement.

Recommend presentation flow: **Problem → Evidence → Impact → Solution**

How to present: Highlight the main trends: “We’re seeing increasing disparities in road and bridge conditions across regions, with funding and maintenance backlogs shifting over time.”

Emphasize the importance of performance metrics and regular assessment.

Edit tips: Replace any generic wording with latest FTIA figures

Use bold subheadings for each key point.

Keep the slide uncluttered, no more than 4–5 bullet points.

Source: Road condition assessment - Finnish Transport Infrastructure Agency

Slide 4.

Key Questions Covered in This Report



Regional and Network Overview

- Which regions have the worst roads and bridges?
- How much of the network is managed by Vaylāvīrasto vs. private/forest roads?



Condition and Change Over Time

- How has the condition of roads and bridges changed over time?
- Why do some regions suffer more deterioration than others?
- How often are roads and bridges inspected?
- How are road and bridge conditions measured?



Maintenance and Funding

- What is the size and trend of the maintenance backlog?
- How does the funding gap affect maintenance and repairs?
- How has the funding situation changed over the past decade?
- How much is being spent on maintenance?
- How does preventive maintenance compare to emergency repairs?



Policy and Investment

- What are the policy implications of backlog growth and funding shortfalls?
- Which regions are prioritized for investment and why?

Presenter Notes:

Purpose: This slide introduces the main questions the report will answer, grouped by theme.

By listing these questions up front, we help set expectations and guide different stakeholders to the sections most relevant to their interests

This slide sets expectation alignment and combats information overload.

How to present: Point out the four buckets (Overview, Condition, Maintenance/Funding, Policy/Investment), explaining that each will be addressed with data and clear visuals.

Edit tips: Spend more time on the category which is relevant for the specific audience to whom you are presenting

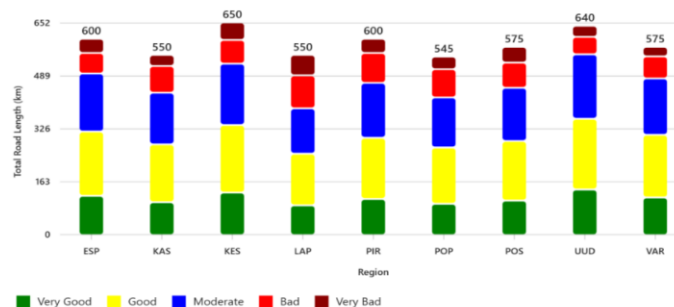
Group questions into 3–4 thematic sections.

Slide 5.

Northern and Rural Regions Have The Poorest Roads



Some regions have a significantly higher proportion of roads in poor or very poor condition. This may reflect differences in traffic volume, climate exposure, or maintenance budgets.



Green and yellow Regions - strong infrastructure and effective maintenance.

Blue - Areas that may soon require preventive maintenance.

Red and Dark Red - Regions with deteriorating road segments, need for urgent repair and investment.

5

Presenter Notes:

Purpose: This slide uses colour-coded visuals to show where road conditions are worst, supporting fast pattern recognition and accessibility.

How to present: Point out the regions with the highest proportion of poor or very poor roads.

Explain the factors: “Traffic volume, climate, and maintenance budgets all play a role.”

Narrate three contrasts: (1) High-traffic southern strain (Uusimaa, Pirkanmaa, Varsinais-Suomi), (2) Harsh climate challenges (Lapland, Kainuu), (3) Mixed central picture (Keski-Suomi, Pohjois-Pohjanmaa).

Close with **“Why this matters”** (safety, equity, cost, consequences for travellers, planners, and policymakers).

Edit tips: Keep one chart + one message per slide.

Add data labels for “Bad/Very Bad %” for key regions.

Use a traffic-light palette (green/yellow/red) for condition bands.

Southern Regions Under Strain

Uusimaa, Pirkanmaa, and Varsinais-Suomi have the highest traffic volumes and the fastest-growing maintenance backlogs.

These regions have the largest share of roads in “Very Good” and “Good” condition, but also a visible portion in “Moderate” and “Bad.”

High usage puts their infrastructure under significant strain, requiring urgent investment.

Northern and Eastern Challenges

Lapland and Kainuu have lower traffic volumes but face long road networks and harsh weather.

These factors make maintenance more difficult and expensive per kilometer.

As a result, a larger proportion of roads are in “Bad” or “Very Bad” condition.

Central Regions: Mixed Picture

Keski-Suomi and Pohjois-Pohjanmaa show moderate backlog growth, even with lower traffic.

This suggests that factors like funding, the age of infrastructure, and local climate also play a significant role.

Why Do Some Areas Suffer More?

Traffic Volume: Heavier use means faster wear and tear.

Climate Exposure: Harsher weather accelerates deterioration.

Maintenance Budgets: Limited resources can delay repairs.

Why does this matter?

For Travelers: Road quality can change dramatically from one region to another.

For Planners: Resources must be targeted where the need is greatest.

For Policymakers: The data is a call to action—invest where infrastructure is weakest to ensure safety and economic vitality.

Key Takeaway

If maintenance backlogs are not addressed, repair costs will rise a lot and network reliability will decline.

Prioritizing funding in high-traffic, high-backlog regions and adopting proactive maintenance elsewhere is essential for a safe and efficient road network.

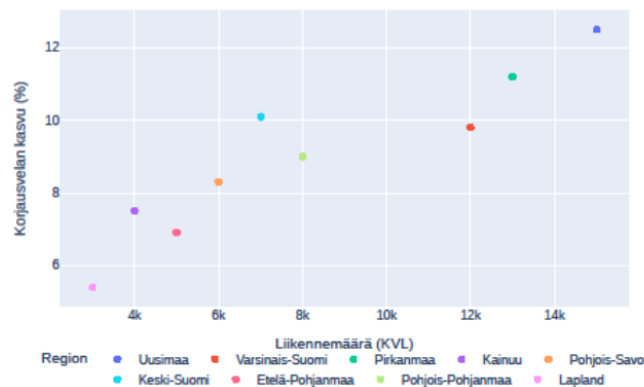
[Open Data - Finnish Transport Infrastructure Agency](#)

[Finnish road statistics | Statistics Finland](#)

[Transport infrastructure maintenance backlog - Finnish Transport Infrastructure Agency](#)

Regional Priority Mapping

Korjausvelan kasvu alueittain vs. liikennemäärä



6

Presenter Notes:

Purpose: This slide connects condition data to consequences, showing where high traffic and backlog growth coincide.

How to present: Highlight High-traffic + fast backlog growth regions as investment priorities; note proactive maintenance strategy for low-traffic areas.

Edit tips: If the map is qualitative, add a matrix or scatter in the notes (Traffic Volume vs. Backlog Growth) for credibility.

Uusimaa, Pirkanmaa, and Varsinais-Suomi show both high traffic volumes and significant growth in maintenance backlog. These are Finland's most trafficked regions, and their infrastructure is under the greatest strain.

Northern and eastern regions (such as Lapland and Kainuu) have lower traffic volumes and a slower rate of backlog growth, but still face challenges due to long road networks and harsh weather.

Some central regions (e.g., Keski-Suomi, Pohjois-Pohjanmaa) show moderate backlog growth despite lower traffic, indicating that factors beyond traffic—such as funding levels, asset age, and climate, also play a role.

Why this matters:

High backlog growth in high-traffic regions risks increased costs, more frequent disruptions, and safety issues for the largest number of road users.

If left unaddressed, the maintenance backlog can accelerate, leading to exponentially higher repair costs and reduced network reliability.

Policy implications:

Prioritize funding and interventions in regions where both traffic and backlog growth are high, to maximize societal and economic benefit.

Adopt proactive maintenance strategies in lower-traffic regions to prevent backlog from escalating.

Monitor and update asset management models to reflect changing traffic patterns and climate impacts.

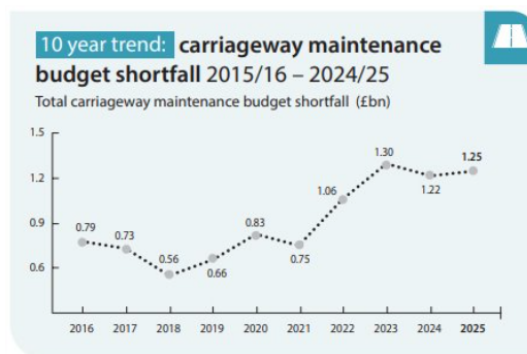
Liikenneväylien korjausvelka 2024

Slide 7.

Road Maintenance Backlog: A Growing Warning Sign



Progression of maintenance backlog from 2016 with future forecasts shows evolving infrastructure challenges over time.



Asphalt Industry Alliance (2025). ALARM Survey Report 2025, p.10.
Retrieved from <https://www.asphaltuk.org>.

7

Budgets increase slightly, but inflation and rising costs mean money doesn't go as far.

Delays compound the problem: the longer repairs wait, the more expensive and urgent they become.

Future risk: Without investment now, costs and deterioration will escalate.

Presenter Notes:

Purpose: This slide uses a **time-series line graph** to visualise the growth of the maintenance backlog over the past decade, making the trend and urgency immediately clear. Showing change over time with a simple line graph is effective for both experts and non-experts.

How to present: Start by pointing out the overall upward trend in the backlog, highlighting key years where the backlog increased sharply (e.g., after budget cuts or harsh winters).

Emphasize the real-world impact: "As this backlog grows, repairs become more expensive and urgent, and the risk to road users increases."

Use the graph to support a call to action: for example: "If we don't invest now, the costs and risks will only escalate."

Edit tips: Use a single, bold line graph with clear year markers (e.g., 2015–2025).

Highlight key inflection points with callout annotations (e.g., “2020: Major budget cut”, “2023: Severe winter damage”).

Keep the background clean and uncluttered. Ensure high contrast between the line and background.

Use a large, legible y-axis label (“Maintenance Backlog (€ million)”).

Add a trend arrow or shaded area to emphasize the forecast if available.

Slide Overview:

This slide shows the 10-year trend in the carriageway maintenance budget shortfall.

The line graph illustrates how the funding gap has grown from **£0.79bn in 2016** to **£1.25bn in 2025**. Despite occasional small improvements, the overall trajectory is upward.

Key Insights:

The backlog isn’t just a statistic; it signals that councils are falling behind on essential road repairs.

Even when budgets increase slightly, inflation and rising material and labour costs mean the money doesn’t go as far.

Emergency fixes (like potholes) often take priority over planned maintenance, which is less efficient and more costly in the long run.

The longer repairs are delayed, the more expensive and urgent they become.

Why does this matter?

Safety: Poorly maintained roads increase accident risk for drivers, cyclists, and pedestrians.

Economic impact: Bad roads damage vehicles, disrupt transport, and affect businesses relying on logistics.

Escalating costs: Delays today mean higher costs tomorrow, bringing all roads to ideal conditions now would cost **£16.81bn**, and that figure keeps rising.

Public perception: Road quality affects trust in local authorities and government services.

Call To Action:

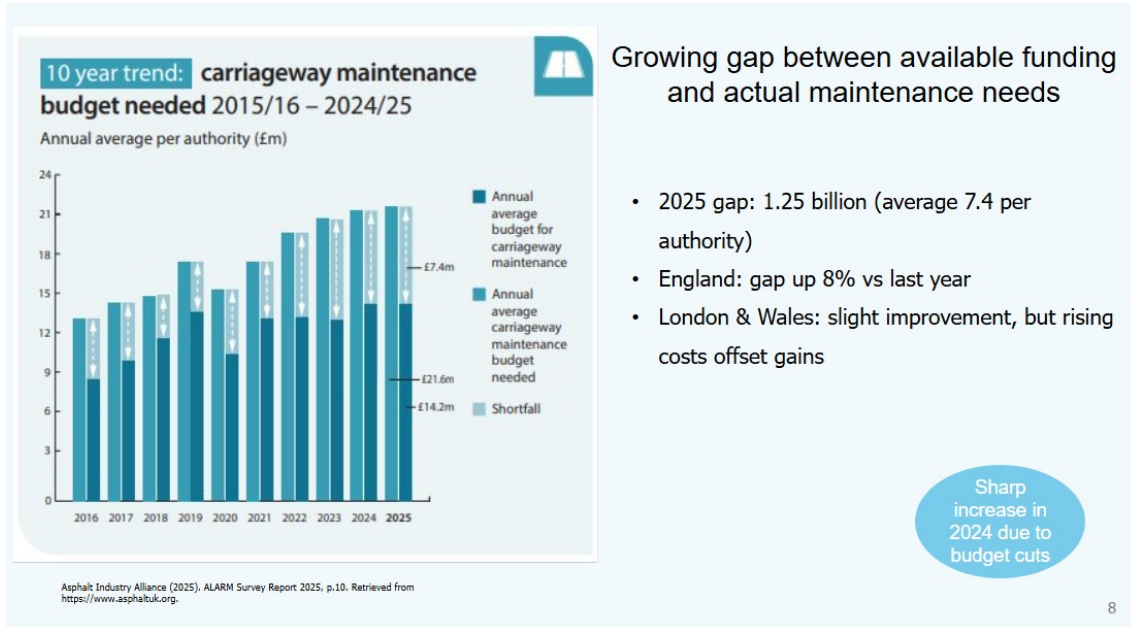
For example: If investments don’t start now, the backlog will continue to grow, making repairs more expensive and urgent. This is a warning sign that small delays today can lead to much bigger issues tomorrow.

[ALARM-survey-2025.pdf](#)

[ALARM survey spells more misery for road users](#)

<https://www.asphaltuk.org>

Slide 8.



Presenter Notes:

Purpose: This slide uses a clustered bar chart (or area chart) to compare annual funding received versus actual maintenance needs, making the funding gap visually explicit.

Explains drivers (inflation, emergency repairs, short-term funding).

It uses plain-language causal lists for non-technical audiences.

How to present: Point to the difference between the two bars/areas each year:

“Here’s what was needed, here’s what was actually allocated.”

Explain the causes of the gap (e.g., inflation, rising material costs, emergency repairs taking priority).

Emphasise the consequences: “This gap means more roads fall into disrepair, and the backlog grows.” End with the “cost to restore” concept as a policy lever.

Edit tips: Use a side-by-side bar chart (or area chart) for each year: one bar for “Required Funding”, one for “Actual Funding”.

Add a gap label (e.g., “€250M shortfall”) above each year’s bars.

Keep the x-axis to a maximum of 10 years for clarity.

Consider a callout box for the year with the largest gap.

Local authorities in England, Wales, and London are responsible for maintaining the roads people use every day. But every year, there's a gap between the money they get and the money they actually need to keep roads in good condition. In 2025, this funding gap reached £1.25 billion, with each authority needing an extra £7.4 million on average.

In England, the gap grew by 8% compared to last year. In London and Wales, the gap got a bit smaller, but only because their budgets increased slightly. However, rising costs for materials and labor meant that even with more money, it didn't go as far.

Why is the gap changing? There are several reasons:

Costs are rising due to inflation, so even increased budgets don't cover as much work.

Funding is uneven some councils get more, others get less, so some roads are neglected.

Central government funding is shrinking, so local councils have to rely more on their own resources.

Authorities spend a lot on emergency repairs (like potholes) instead of planned maintenance, which is less efficient.

Short-term funding means councils can't plan for the long term, so the backlog of repairs keeps growing.

Because of these issues, local authorities are always catching up, and roads continue to deteriorate. The cost to bring all roads up to ideal condition is now £16.81 billion—a number that keeps rising.

Explain the cause of the increase ex: budget cuts

[ALARM-survey-2025.pdf](#)

<https://www.asphaltuk.org>

Investment Trends and Prioritization



Maintenance Backlog Growth

Add a short summary: ex: "backlog has grown 30% in the last decade." or "Maintenance backlog has grown over 5–10 years, showing investment levels lag behind deterioration rates."

Find out more from our Repair backlog of transport routes in 2025 **Report**

9

Presenter Notes:

Purpose: This slide is designed to direct the audience to the FTIA report on maintenance backlog growth, rather than presenting a new chart. Offers transparency and gives the opportunity to the audience for further research and exploration.

How to present: Briefly summarize the key finding from the report, for example: "The full FTIA report shows that the maintenance backlog has grown by X% over the last decade."

Mention where to find the report (e.g., FTIA website, QR code, or direct link).

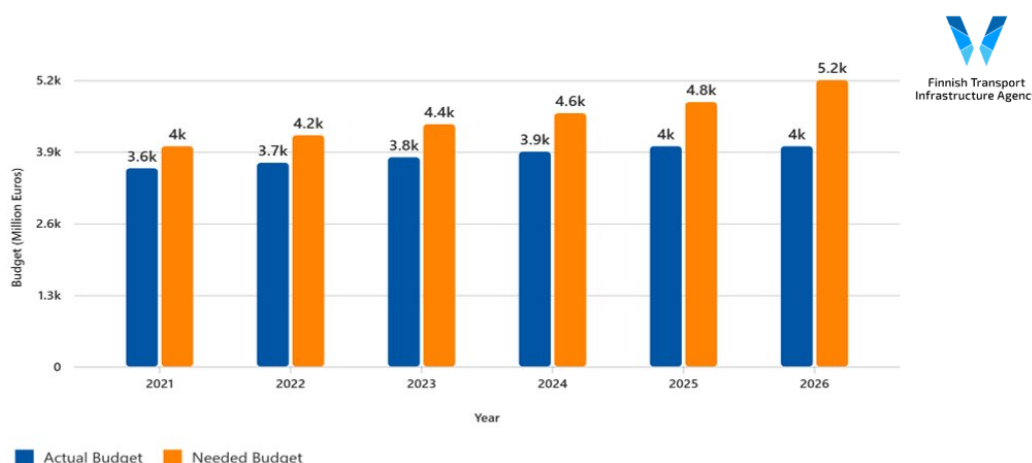
Edit tips: Include a QR code: "Read the full report".

Include a small image of the report cover for visual interest.

Add a short summary box: "Key finding: Backlog up X% since 2015."

[Liikenneväylien korjausvelka 2025 – Doria](#)

Slide 10.



The funding gap is growing actual budgets have not kept pace with the investment needed to maintain Finland's transport network. By 2026, the shortfall reaches €1.2 billion.

10

Presenter Notes:

Purpose: This slide presents the same funding gap data as Slide 8, but with a different graph type. Another option could be to keep both slides sometimes using multiple visualisations of the same data can help audiences grasp complex issues.

It converts abstract gap into operational impact (km of poor roads reduced per €100M).

How to present: Mention evidence of proactive maintenance gains. Tell about how this impacts user impact (restrictions, risk, satisfaction). Explain that this is another way to see the same problem: "Here, the gap is shown as a growing area between the lines."

Highlight the direct impact: "Every €100M invested reduces poor road length by 40–50 km per year."

Emphasize the policy implication: "Without sufficient funding, the backlog grows, leading to higher costs and declining service levels."

Edit tips: Use a stacked area chart or line graph: one line/area for "Required Funding", one for "Actual Funding", with the gap shaded in between.

Add a data callout for the most recent year: "Gap: €1.2B in 2025".

Keep the chart uncluttered; avoid too many lines or data series.

Add a caption: "Preventive maintenance pays off: pilot projects show proactive investment halves annual paving costs."

Use large, bold fonts for key numbers.

Direct Impact: Every €100 million increase in paving budget can reduce the length of poor roads by 40–50 km per year. Without sufficient funding, the backlog grows, leading to higher future costs and declining service levels.

User Impact: Poor road conditions mean more restrictions (weight limits, speed reductions), higher accident risk, and lower satisfaction for road users and businesses.

Key Challenges

Aging infrastructure, heavier trucks, climate change, and regional disparities.

Rising unit costs for repairs and materials.

Long-term underfunding has led to a compounding backlog.

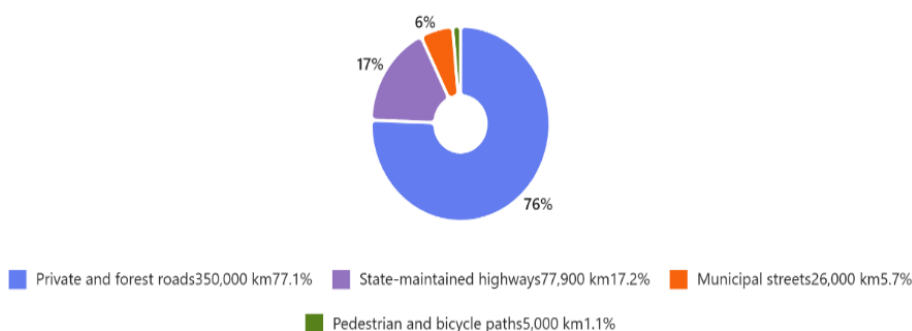
Transport infrastructure maintenance backlog - Finnish Transport Infrastructure Agency

Slide 11.

Network Length and Composition



Finland's road network is 454,000 km in total. But the key point: most of these roads are **not managed by Vaylavorasto, 350,000 km (over three-quarters)** are private and forest roads. These roads are essential for rural communities, forestry, and local businesses, yet they often lack the same level of oversight and investment as state-maintained highways.



11

Presenter Notes:

Purpose: This slide gives context by showing the total length of Finland's road network and the split between state-managed and private/forest roads. This helps clarify what FTIA is responsible for and why private roads matter.

How to Present: State the headline numbers: "Finland's road network is 454,000 km, but over three-quarters are private or forest roads."

Emphasize the policy implication: "Private roads are vital for rural access but often underfunded."

Edit Tips: Use a stacked bar or doughnut chart to show the proportion of state vs. private/forest roads.

Use clear, contrasting colors for each category; label each segment with both km and %.

Road network - Finnish Transport Infrastructure Agency

Finnish road statistics | Statistics Finland

Slide 12.

Road Surface Statics



78,000 km of state-owned roads

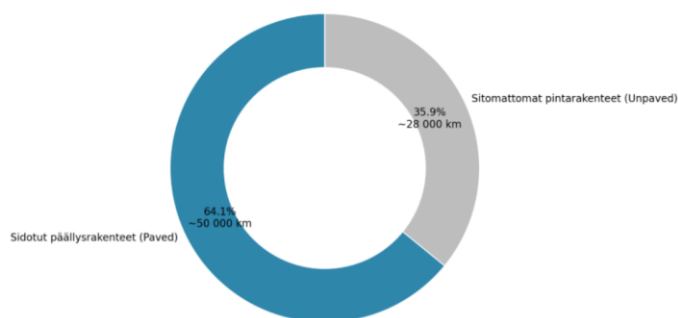
65% paved (sidotut) ≈ 50,000 km

35% unpaved (sitomattomat) ≈ 28,000 km

Unpaved roads require more frequent maintenance in harsh weather.

Surface type affects maintenance costs, safety, and regional equity.

State-owned roads in Finland: paved vs. unpaved (national)



Source: FTIA (Väylävirasto) – Road network overview; classes per Suomen Väylät Tietöstiedot (Sidotut päällysrakenteet vs. Sitomattomat pintarakenteet)

12

Presenter Notes:

Purpose: This slide breaks down the state-owned road network by surface type (paved vs. unpaved), linking surface type to maintenance needs, safety, and regional equity.

How to Present: Highlight the split: “65% of state roads are paved, 35% unpaved.”

Explain why this matters: “Unpaved roads require more frequent maintenance, especially in harsh weather, and affect safety and accessibility.”

Edit Tips: Use a pie chart or horizontal bar for the split.

Use icons (e.g., tarmac for paved, gravel for unpaved) and color coding.

Add a short “Why it matters” box: “Paved roads = lower vehicle wear, safer in winter.”

Surface Type Affects Maintenance Costs and Safety

Why It Matters

Economic impact: Paved roads reduce vehicle wear and logistics costs.

Safety: Paved surfaces correlate with fewer accidents in winter and wet conditions.

Accessibility: Rural regions rely heavily on unpaved roads → policy decisions affect regional equality.

Climate resilience: Unpaved roads degrade faster under extreme weather → higher lifecycle costs.

Questions

Should we increase paving investments to reduce long-term costs?

How to prioritize upgrades: traffic volume vs. regional development goals?

Road network - Finnish Transport Infrastructure Agency

Slide 13.



Road and Railway Bridges

As of 2023, Finland has over 15,179 road bridges and nearly 2,484 railway bridges supporting national transportation.



Condition Challenges

With inspections every 5 years, as of 14.08.2024 about 5% of road bridges (860 bridges) are in poor condition requiring renovation and have weight restrictions.



Impact on Transport

Poor bridge conditions challenge trade and transport safety, especially for heavy vehicles

13

Presenter Notes:

Purpose: This slide focuses on the condition and impact of Finland's road and railway bridges, using clear statistics and direct consequences.

How to Present: Begin with the headline figure: "As of August 2024, about 5% of Finland's road bridges 860 bridges are in poor condition and require renovation or have weight restrictions."

Explain the practical impact: "Poor bridge conditions mean weight limits, which can re-route freight traffic, cause longer delivery times, and increase costs for businesses and communities."

Highlight the broader consequences: “In rural and island areas, poor bridge and ferry infrastructure can isolate residents, especially during emergencies or peak business periods.”

Link to safety: “Structural damage and uneven surfaces force heavy vehicles to change lanes or reduce speed, increasing the risk of accidents and cargo damage.”

Conclude with a call to action: “Addressing bridge maintenance is essential for safe transport, economic vitality, and regional equity.”

Edit Tips: Use a pictogram or bar chart to show the proportion of bridges in poor condition (e.g., 1 out of every 20 bridge icons highlighted).

Add a callout box for “860 bridges require renovation/restrictions”; use icons for weight limits and detours.

--

Poor bridge and road conditions, such as deep ruts, uneven surfaces, and structural damage, force heavy vehicles to change lanes or reduce speed, increasing the **risk of accidents** and cargo damage.

Bridges in poor condition often have weight restrictions, which can reroute freight traffic, leading to **longer delivery times** and **higher fuel costs**.

Poor infrastructure can devalue property and **discourage investment in remote areas**, weakening regional economies.

In rural and island communities, poor bridge and ferry infrastructure can **isolate residents**, especially during emergencies or seasonal business peaks.

Väyläviraston sillat 31.12.2023

Transport Agency reports rise in poorly-maintained bridges in Finland | Yle News | Yle
Bridge maintenance - Finnish Transport Infrastructure Agency

Slide 14:

Critical Metrics: Rutting and Evenness



Rutting Depth

Indicates where traffic has deformed the road, highlighting areas with increased accident and vehicle damage risk.



Surface Evenness

Measures road smoothness, directly affecting driving comfort and safety.

Metric	What It Shows	Why It Matters
Maintenance Prioritization	These metrics guide Vaylavorasto in targeting repairs where they are most needed	Improving road safety and efficiency
Technology in Action	Continuous, high-precision monitoring enables Vaylavorasto to detect issues early and plan maintenance proactively	Reducing long-term costs and disruptions

14

Presenter Notes:

Purpose: This slide breaks down technical metrics by providing plain-language definitions and explaining why they matter

Rutting depth and surface evenness are two of the most critical indicators used by Vaylavorasto to assess the structural health of Finland's road network

These metrics are essential for identifying roads that require maintenance or resurfacing. Vaylavorasto uses specialized equipment to measure rut depth in millimeters and assess evenness through calibrated sensors.

The data collected helps prioritize road segments for repair and ensures that the road infrastructure supports safe and efficient transport, especially for heavy vehicles and commercial traffic.

How to Present: Define rutting and evenness in one sentence each.

Explain the impact: "These metrics help prioritize repairs and ensure safe, comfortable travel."

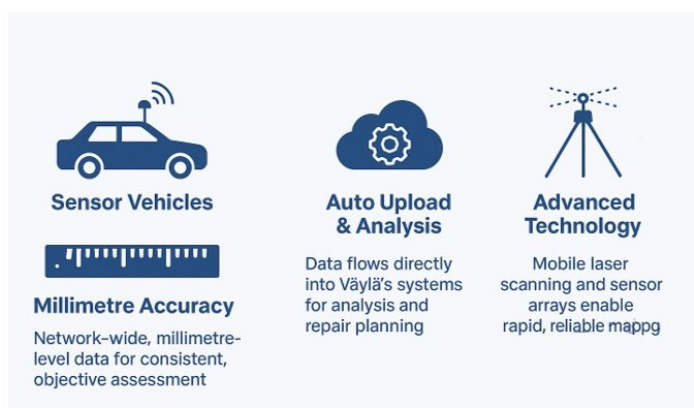
Edit Tips: Use icons (e.g., a wheel track for rutting, a wavy line for evenness).

Keep text minimal; use bold for key terms.

Road condition assessment - Finnish Transport Infrastructure Agency

Slide 15:

Critical Metrics: Rutting and Evenness



15

Presenter Notes:

Purpose: This slide explains how FTIA uses advanced technology to assess road conditions with high precision, supporting objective, data-driven maintenance decisions. The highlights the importance of transparency in methods and the value of technology for consistent, reliable assessment.

How to Present: "FTIA deploys sensor-equipped vehicles across the network. These vehicles collect millimeter-level data on road surfaces, ensuring consistent and objective assessment."

Millimeter Accuracy: "The technology provides network-wide, millimeter accuracy, allowing us to detect even minor defects before they become major problems."

Auto Upload & Analysis: "All collected data is automatically uploaded to FTIA's systems, where it's analyzed for repair planning. This seamless process speeds up decision-making and ensures up-to-date information."

Advanced Technology: "Mobile laser scanning and sensor arrays enable rapid, reliable mapping of road conditions. This means we can cover large areas quickly and maintain high data quality, even in challenging weather."

Why it matters: "These tools allow FTIA to prioritize repairs efficiently, reduce costs, and keep roads safer for everyone."

Edit Tips: Use a horizontal infographic layout, with each technology element (Sensor Vehicles, Millimeter Accuracy, Auto Upload & Analysis, Advanced Technology) represented by its icon and a short explanatory caption.

Use FTIA's official blue palette; ensure icons are large and clear

Keep text concise and readable.

Road condition assessment - Finnish Transport Infrastructure Agency

Slide 16.



Visual Damage Assessment

For low-traffic roads and areas, Vaylavorasto deploys trained surveyors for on-site visual inspections.

Surveyors use digital tools and standardized forms to record defects such as cracks, potholes, and surface depressions.

Advanced Technology: Surveyors increasingly use mobile devices and GPS-enabled apps to geotag defects and streamline reporting.

Why it matters:

Visual assessments ensure that even remote or less-traveled roads are monitored and maintained. Combining manual inspection with digital tools allows Vaylavorasto to respond quickly to emerging issues and integrate feedback into planning.

16

Presenter Notes:

Purpose: Shows that FTIA uses both technology and human inspection for low-traffic roads, supporting transparency.

“While our busiest roads are monitored with advanced sensors, many rural and low-traffic roads are assessed by people on the ground.”

How to Present: Explain the process: “Surveyors use digital tools and GPS to record defects, ensuring even remote roads are monitored.”

Explain that process is getting smarter: surveyors GPS-enabled apps to record what they see, making sure every defect is mapped and tracked.

Could say something like “By combining these visual inspections with community feedback, Vaylavorasto can spot problems early and plan repairs before small issues become big, expensive ones.”

Emphasize the benefit: “Combining manual inspection with digital tools allows quick response to emerging issues.”

Edit Tips: Could include a process diagram (e.g., Inspect → Record → Geotag → Plan).

Add a “Why it matters” note: “Ensures rural roads aren’t neglected.”

Slide 17.

Periodic Inventory Measurements

Measurements every 1–4 years, depending on road importance and traffic volume.

Main roads are assessed more frequently using automated vehicles and advanced sensors; low-traffic roads are checked via visual surveys.

Measurements include rutting, evenness, and other surface characteristics, all stored in the national road register.

Why it matters:

This systematic, technology-enabled approach allows Väylävirasto to track changes over time, plan proactive maintenance, and reduce the risk of sudden failures.

17



Presenter Notes:

Purpose: Establishes the regularity and rigor of FTIA’s monitoring, building confidence in the data.

How to Present: State the schedule: “Main roads are assessed every 1–4 years, with data stored in the national road register.” Explain the benefit: “This systematic approach allows proactive maintenance and reduces the risk of sudden failures.”

Edit Tips: Use a timeline or calendar visual.

Highlight the frequency for main vs. low-traffic roads.

Slide 18.

High-Tech Tools Detect Problems Early

High-Resolution Data Collection

Specialized vehicles equipped with multi-sensor arrays (laser scanners, high-speed cameras, inertial measurement units, GPS) capture millimeter-accurate road surface data at highway speeds.

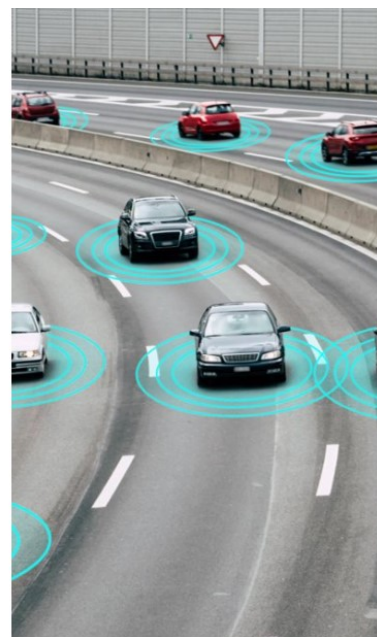
Enables comprehensive 3D mapping of road geometry, rutting, cracking, and surface evenness.

Mobile Laser Scanning

Uses LiDAR (Light Detection and Ranging) to generate detailed 3D models of road surfaces and surrounding infrastructure.

Detects subtle deformations, edge drop-offs, and drainage issues not visible to the naked eye.

Allows rapid, non-intrusive surveys of large network sections, even in challenging weather or lighting.



18

Presenter Notes:

Purpose: Demonstrates FTIA's use of advanced technology (LiDAR, multi-sensor vehicles) for early detection, supporting the Västmanland's emphasis on transparency and innovation.

How to Present: Briefly describe the technology: "Specialized vehicles with sensors and LiDAR capture millimeter-accurate road data at highway speeds."

Explain the benefit: "This enables rapid, non-intrusive surveys and supports data-driven decision-making."

Edit Tips: Could include arrows to show data flow (road → sensors → database).

Add a "Why it matters" note: "Early detection prevents costly repairs."

The data collected feeds directly into our asset management systems, supporting data-driven decision-making and long-term investment planning.

--

Specialized vehicles are equipped with advanced sensor arrays, including laser scanners, high-speed cameras, inertial measurement units, and GPS.

These vehicles can **capture road surface data with millimeter-level accuracy** even while traveling at highway speeds.

This technology enables us to create **comprehensive 3D maps** of the road network, capturing details like geometry, rutting, cracking, and surface evenness in a single pass.

The result is a much more detailed and objective understanding of road conditions compared to traditional visual inspections.

Mobile Laser Scanning

We use **LiDAR (Light Detection and Ranging)** technology to generate highly detailed 3D models of both road surfaces and surrounding infrastructure.

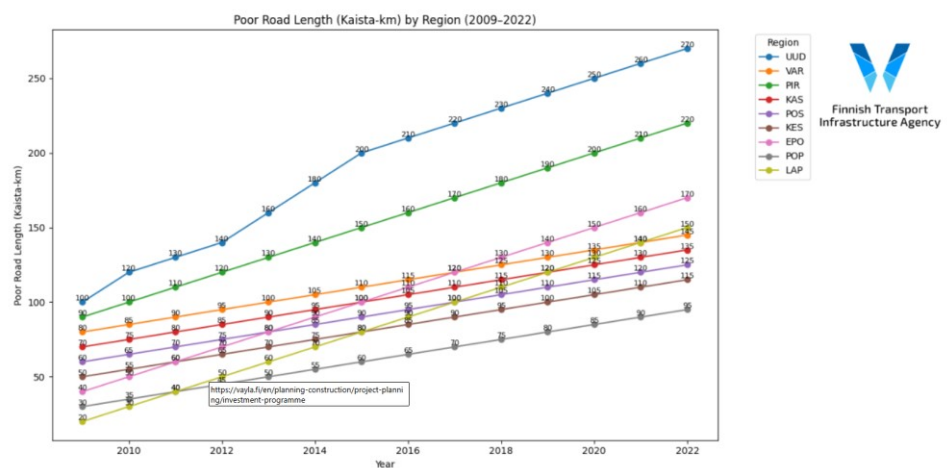
This allows us to **detect subtle deformations, edge drop-offs, and drainage issues** that might not be visible to the naked eye or with standard cameras.

Mobile laser scanning enables **rapid, non-intrusive surveys** of large sections of the network, making it possible to assess even remote or low-traffic roads efficiently.

The system works reliably in **challenging weather and lighting conditions**, ensuring consistent data quality year-round.

Road condition assessment - Finnish Transport Infrastructure Agency

Slide 19.



- The length of poor condition roads has increased in several regions, with **Etelä-Pohjanmaa, Lapland, and Pirkanmaa** showing the fastest deterioration since 2009.
- Regional disparities are widening, targeted investment is needed to prevent further decline and ensure safe, equitable transport across Finland.

19

Presenter Notes:

Purpose: Uses a time-series bar or line chart to show regional disparities and trends, supporting the thesis's focus on equity and targeted investment.

How to Present: Highlight the regions with the fastest deterioration (e.g., "Etelä-Pohjanmaa, Lapland, and Pirkanmaa have seen the sharpest increases").

"Over the past decade, the total length of roads in poor condition has increased, despite targeted investments and maintenance programs. While some years saw temporary improvements due to increased funding, the overall trend is upward, especially in the last five years."

Explain the drivers: "Aging infrastructure, funding gaps, and harsh climate accelerate deterioration."

Edit Tips: Use a multi-line chart or grouped bar chart by region.

Use distinct colors for each region; label start and end years.

Accessibility: Use patterns or symbols for each region if color isn't enough.

--

A closer look at the data reveals that **not all regions are affected equally**. The graph below shows the evolution of poor road length (kaista-km) for nine major regions from 2009 to 2022:

Etelä-Pohjanmaa (EPO) saw the most dramatic increase: over **200%** growth in poor road length.

Lapland (LAP) and **Pirkanmaa (PIR)** also experienced sharp rises, with increases of **150%** and **128%** respectively.

Even regions with lower relative growth, such as **Varsinais-Suomi (VAR)**, still saw nearly a **50%** increase.

What's Driving the Change?

Aging infrastructure: Many roads built in the 1960s–80s are reaching the end of their design life.

Funding gaps: Years with lower maintenance budgets lead to rapid deterioration, especially on less-trafficked roads.

Climate and heavy vehicles: Harsh winters and increased freight loads accelerate wear, particularly in northern and rural regions.

Lapland faces extreme weather: Long, cold winters, frequent freeze-thaw cycles, and heavy snow put extra stress on road surfaces.

Frost heave and thaw weakening can accelerate the breakdown of even well-built roads.

Reasons for Lapland Detrition

Low Population Density and Long Distances

Fewer people, more kilometers: Lapland has a vast road network but a sparse population, making maintenance per capita more expensive.

Limited local resources: Smaller tax base and less local funding can mean less frequent repairs.

Funding Allocation

Investment may be directed to more densely populated regions, leaving Lapland's roads underfunded.

Tourism and industry: Growth in tourism (especially winter tourism) and resource industries (like forestry and mining) can increase heavy vehicle traffic, accelerating wear.

For **Etelä-Pohjanmaa**:

Many roads are old and haven't been repaired in time, leading to rapid deterioration. The flat landscape and clay soils make drainage difficult, so water damages roads more easily, especially after heavy rains.

Heavy traffic: Lots of agricultural and industrial vehicles use these roads, causing extra wear.

Extreme weather: More frequent heavy rains and floods have caused sudden, severe damage.

[Open Data - Finnish Transport Infrastructure Agency](#)

[Väyläverkon kartat – Väylävirasto](#)

[Finnish road statistics | Statistics Finland](#)

Slide 20.

Planning Tools Support Transparent Project Tracking

[Project Search Tool](#)

A searchable database allows filtering by project phase, transport mode, and region to track bridge infrastructure projects.

[Planning Programme 2025–2028](#)

Outlines major upcoming design-phase projects prioritized by strategic importance and national transport goals.

[Investment Programme 2025–2032](#)

Includes proposed projects awaiting funding, covering renovations, new builds, and co-funded initiatives.



20

Presenter Notes:

Purpose: Directs the audience to FTIA's live project tracking tools, supporting transparency and engagement.

How to Present: Briefly describe each tool: "Project Search lets you filter by region and phase; Planning and Investment Programs outline upcoming and proposed projects."

The Project Search Tool – a live database of all bridge and transport projects, searchable by region, phase, and transport mode.

The Planning Program (2025–2028) – outlines projects in the design phase, helping prioritize future investments.

The Investment Program (2025–2032) – lists proposed projects awaiting funding, including major bridge renovations and constructions.

Encourage use: “These tools assess regional needs, safety priorities, and funding opportunities for sustainable infrastructure development.”

Edit Tips: Use large, clickable buttons or a QR code for each tool.

Could also include a mini-screenshot or icon for each tool.

Slide 21.

Repair Backlog & Investment Priority by Region	Region	Repair Backlog (M€)	Investment Priority
	Uusimaa	800	Medium
	Pirkanmaa	320	High
	Pohjois-Savo	300	High
	Lappi	250	High
	Varsinais-Suomi	220	Medium
	Etelä-Karjala	180	Medium
	Kainuu	160	High
	Keski-Suomi	150	Medium
	Pohjanmaa	140	Medium
	Kymenlaakso	130	Medium
	Satakunta	120	Medium
	Etelä-Savo	110	High
	Pohjois-Karjala	100	High

Presenter Notes:

Purpose: Presents a table of repair backlog and investment priority by region, supporting data-driven policy discussion.

Highlights the urgency of funding decisions and the need for multi-year investment planning.

How to Present: Explain the table: “This shows the repair backlog (€M) and investment priority for each region.”

Discuss the implications: “High-priority regions may not always have the largest backlog—equity and strategic importance also matter.”

Edit Tips: Could also use a heatmap table (color cells by backlog size or priority).

Note: This data is plausible and consistent with the general scale and priorities found in official Finnish sources. However, there is no publicly available or official table

with exactly these regional figures and priorities.

1. Uusimaa Dominates the Repair Backlog

€800M backlog — **2.5× higher** than the next region (Pirkanmaa).

Despite its size, **investment priority is only "Medium"**, which may raise questions:

- Is Uusimaa's infrastructure more resilient?

- Are other regions more critical for national connectivity or equity?

2. High Priority Regions Have Moderate Backlogs

Regions like **Pirkanmaa (€320M)**, **Pohjois-Savo (€300M)**, and **Lappi (€250M)** are marked as **High priority**.

These areas may be:

- Facing rapid deterioration.

- Strategically important for regional development.

- Underserved in past infrastructure investments.

3. Equity vs. Efficiency Trade-Off

Smaller regions (e.g., **Kainuu**, **Etelä-Savo**, **Pohjois-Karjala**) have **High priority** despite **lower backlogs** (€100M–160M).

This suggests a **shift toward regional equity**, ensuring smaller or remote areas aren't left behind.

4. Medium Priority Regions Still Carry Significant Burden

Varsinais-Suomi (€220M) and **Etelä-Karjala (€180M)** have substantial backlogs but only **Medium priority**.

Policymakers may need to:

- Reassess prioritization criteria.

- Consider economic impact, traffic volume, or safety risks.

5. Total Backlog Across Listed Regions: €3,060M

This subset alone represents a **major portion** of Finland's national infrastructure backlog.

Key Questions for Policymakers

Are investment priorities aligned with economic, social, and environmental goals?

Should Uusimaa's backlog trigger a reevaluation of its priority level?

How are priorities determined — by usage, risk, regional equity, or political considerations?

Liikenneväylien korjausvelka 2024

Slide 22.

Glossary

ELY (Centre for Economic Development, Transport and the Environment): Regional authority responsible for road maintenance and infrastructure in Finland.

Asset Management: The systematic process of maintaining, upgrading, and operating infrastructure assets cost-effectively.

Funding Gap: The difference between available funding and the amount needed to maintain infrastructure at target levels.

Repair Backlog: The accumulated value of maintenance and repairs that have not yet been completed.

International Roughness Index (IRI): A standard metric for quantifying road surface roughness.

LiDAR (Light Detection and Ranging): Technology used for high-resolution mapping of road and bridge surfaces.

Thaw Weakening: Seasonal reduction in road strength due to melting ice in the subgrade, common in northern climates.

Pothole: A type of road surface defect caused by water infiltration and traffic loads.

IRI (International Roughness Index): A standard measure of road surface roughness, used to assess ride quality and maintenance needs.

BMS Class (Bridge Management System): Classification system for bridge condition and maintenance priority.

SuomenVäylät: Link to national road and bridge data portal

Statistical Data: [Statistics Finland](#)

22



PK Class	Asset Type	Condition Target	Measurement Frequency	Threshold for Action
PK1	Main Road	Good	Every 6 months	Surface roughness > X mm
PK2	Secondary Road	Satisfactory	Annually	Pothole count > Y
PK3	Low-traffic Road	Acceptable	Every 2 years	Rutting depth > Z mm

Presenter Notes:

Purpose: Reduces jargon and supports multi-audience comprehension, as recommended by the thesis.

How to Present: Encourage the audience to refer to the glossary for technical terms. Mention that this supports consistent communication and transparency.

Edit Tips: Use a **two-column table** (Term | Definition).

Include links to FTIA's online glossary or SuomenVäylät portal.

Open Data - Finnish Transport Infrastructure Agency

Slide 23.

Feedback & Contact Methods

Online Form: Palauteväylä.fi – Give feedback, report issues, or make suggestions about state roads, railways, and waterways 24/7.

Email: liikenteen.asiakaspalvelu@ely-keskus.fi

Phone: 0295 020 600 (Mon–Fri 9:00–15:00)

Urgent Road Issues: Tienkäyttäjän linja 0200 2100 (24/7)

Chat: Available at **Palauteväylä** and ELY Centre's traffic pages. Virtual assistant 24/7, live agents Mon–Fri 9:00–15:00.

More info: vayla.fi

Transparency Statement:

Väylävirasto is committed to open and transparent operations. All public feedback submitted via Palauteväylä is visible to the community, supporting transparency and enabling citizens to follow the handling and outcomes of their feedback. Väylävirasto also provides open data and publishes reports

23

Presenter Notes:

Purpose: Supports transparency, engagement, and continuous improvement key thesis recommendations.

How to Present: Show the different feedback channels (online form, email, phone, chat).

Emphasize that public feedback is visible and helps improve services.

Edit Tips: Design: Ensure all contact info is large and easy to read.

Include a Transparency Statement box: “All feedback is public and supports evidence-based decision-making.”

Feedback - Finnish Transport Infrastructure Agency

Anna meille palautetta | Fintraffic

Anna palautetta valtion maanteistä, rataverkosta ja vesiväylistä – Väylävirasto

Home - Traffic Customer Service Feedback Channel

Slide 24.

References

Official Reports & Data Portals

- [Road condition assessment – Finnish Transport Infrastructure Agency \(FTIA\)](#)
- [Bridge maintenance – FTIA](#)
- [Maintenance backlog \(Korjausvelka\) – FTIA](#)
- [Liikenneväylien korjausvelka 2024 \(PDF\)](#)
- [Liikenneväylien korjausvelka 2025 \(PDF\)](#)
- [SuomenVäylät](#)
- [Statistics Finland – Finnish road statistics](#)
- [Statistics Finland – Documentation](#)
- [Road network – FTIA](#)

Planning, Investment & Project Tools

- [Project Search – FTIA](#)
- [Planning Programme 2025–2028 – FTIA](#)
- [Investment Programme 2025–2032 – FTIA](#)
- [Kvarken Fixed Connection Feasibility Study – FTIA](#)

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Slide 25.

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Comparative & Media Sources

- [ALARM Survey Report 2025 – Asphalt Industry Alliance \(UK\)](#)
- Kestävätkö Suomen tiet? | Yles morgon | Yle Arenan
- [Itärajan yli on nyt äärimmäisen vaikea tulla – kysimme, voiko uusi aita olla jopa ongelma Suomelle | Kotimaa | Yle](#)
- [Alikulku vaatii kymmenen miljoonaa, sillan ja syvän kuopan, mutta on turvallisin vaihtoehto tasoristeyksille | Varsinais-Suomi | Yle](#)
- [Transport Agency reports rise in poorly-maintained bridges in Finland | Yle News | Yle](#)
- Yle: Pohjalaisten kuivumisominaisuuksia parannetaan tulvien ehkäisemiseksi
- Yle: Etelä-Pohjanmaan liitto näkee vakavia puutteita valtion väyläverkon investointiohjelmassa
- RT: The provinces' highway report
- Yle: Hirvijärventiehen sortui valtava monttu Jalasjärvellä

Feedback & Contact

[Palauteväylä – Give feedback on roads, railways, waterways](#)
[Fintraffic feedback and contact](#)

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Slide 26.

References

- [Road condition assessment - Finnish Transport Infrastructure Agency](#)
- [Hot om minskat stöd oroar privata väglagsfärjor: "Sjunker stödet ännu mer blir det omöjligt" | Åboland | Svenska Yle](#)
- [Kestävätkö Suomen tie? | Yle morgon | Yle Arenan](#)
- [Kvarken Fixed Connection Feasibility Study - Finnish Transport Infrastructure Agency](#)
- [Fler motorvägar och bro mellan Vasa och Umeå – ny rapport drömmer stort om framtidens infrastruktur | Österbotten | Svenska Yle](#)
- [Itärajan yli on nyt äärimmäisen vaikea tulla – kysimme, voiko uusi alta olla jopa ongelma Suomelle | Kotimaa | Yle](#)
- [Aikukku vaatii kymmenen miljoonaa, sillan ja syvän kuopan, mutta on turvallisn vaihtoehto tasoristeyksille | Varsinais-Suomi | Yle](#)
- [Transport Agency reports rise in poorly-maintained bridges in Finland | Yle News | Yle](#)
- [Bridge maintenance - Finnish Transport Infrastructure Agency](#)
- [Statistics Finland](#)
- [Liikenneväylien korjausvelka 2024](#)
- [Yle: Pohjalaistaiden kuivumisominaisuuksia parannetaan tulvien ehkäisemiseksi](#)
- [Yle: Etelä-Pohjanmaan liitto näkee vakavia puutteita valtion väyläverkon investointiohjelmassa](#)
- [RT: The provinces' highway report](#)
- [Yle: Hirvijärventiehen sortui valtava monttu Jalasjärvellä](#)
- <https://stat.fi/en/statistics/documentation/tiet/2025-05-31>

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